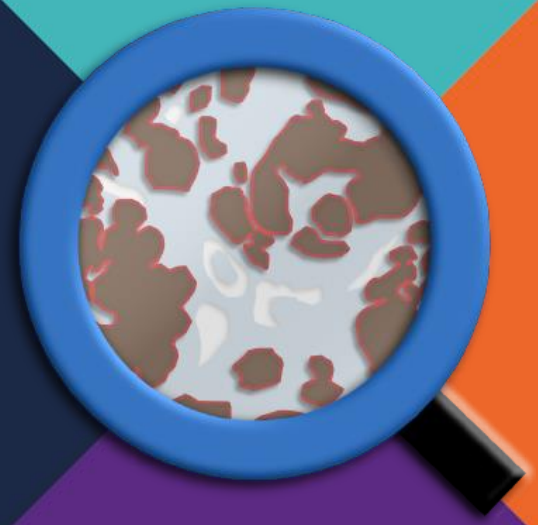
SCIENCE MEETS LIFE

QuPath : a fast Intro

Qupath : Quantitative Pathology & Bioimage Analysis Software



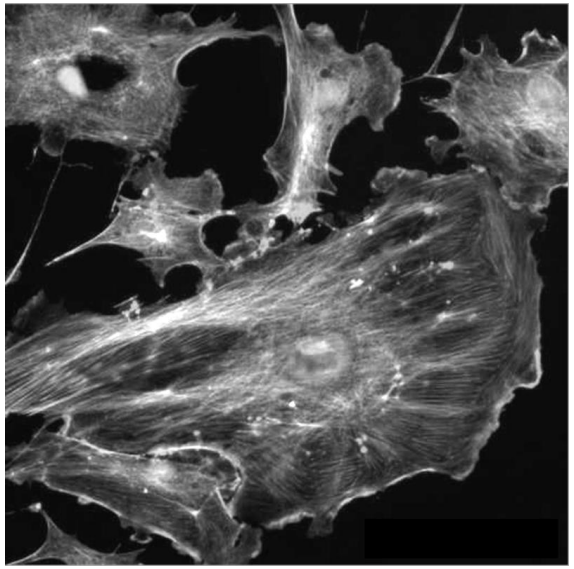
VIB

2026

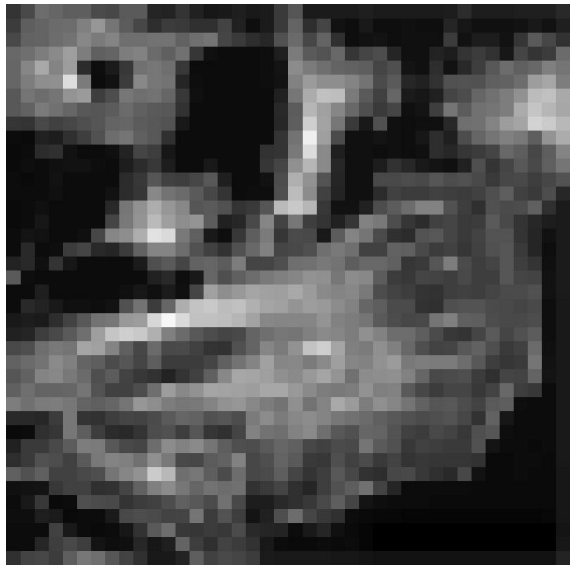
Benjamin PAVIE
Benjamin.pavie@vib.be

Digital Images

Analog Image



Digital Image



Pixel Intensity Values

24	137	142	217	33	43	90	176	124	170	156	66	110	149	49	192
152	221	237	88	184	65	254	2	223	238	208	22	177	241	83	88
120	67	177	113	28	101	57	215	90	206	135	109	65	15	224	106
177	81	148	13	29	18	166	235	114	123	51	65	2	149	120	39
178	30	207	45	163	174	154	196	245	192	115	75	135	72	103	208
162	239	224	169	83	102	98	10	10	106	109	108	71	211	45	159
8	164	252	84	166	250	36	96	248	247	246	30	241	48	247	188
17	122	0	229	191	102	6	179	48	251	158	126	231	112	103	205
81	163	220	30	148	158	107	186	170	220	177	180	100	100	215	17
135	138	156	252	188	39	46	57	149	99	183	62	6	210	156	242
166	165	252	137	59	97	185	68	172	115	88	200	171	172	96	126
103	138	134	180	187	41	94	171	92	62	131	18	213	52	223	192
209	183	122	254	247	193	214	121	158	200	141	100	247	81	200	189
183	133	204	73	221	222	187	159	206	225	39	0	14	34	118	211
247	253	58	105	21	89	145	60	4	232	143	56	114	171	207	39
135	55	127	118	93	174	45	45	21	142	177	0	148	145	229	116
82	26	229	194	94	75	244	211	248	152	108	48	175	43	109	157
26	27	146	208	174	135	67	195	166	37	213	36	183	37	85	237
155	16	215	25	152	212	235	238	58	229	186	68	165	121	152	212

Sampling (Resolution)

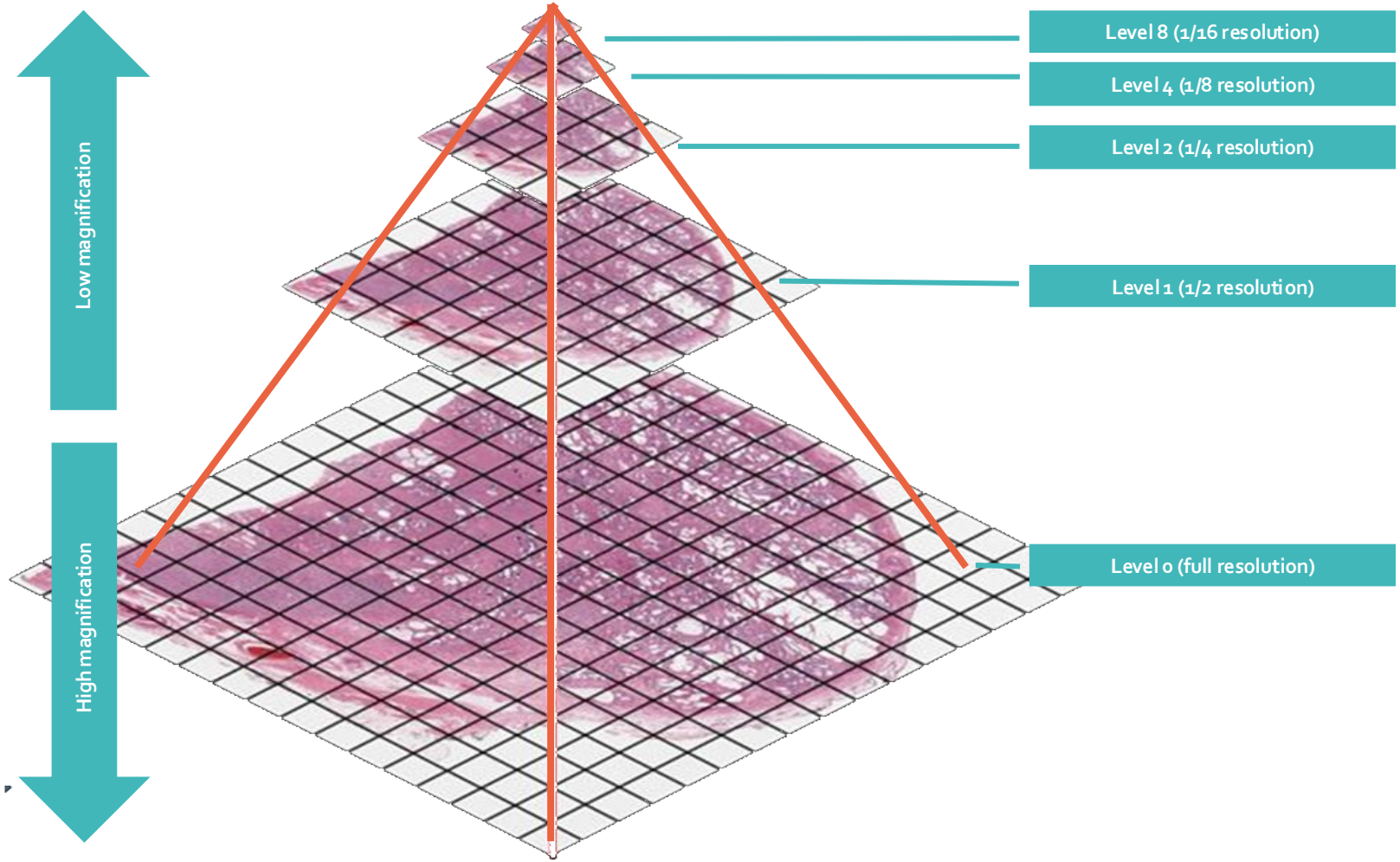
Quantization (Bit-Depth)

VIB

SCIENCE MEETS LIFE

Virtual slides image

Pyramidal Images



VIB

SCIENCE MEETS LIFE

QuPath

Installation

- Installer : https://github.com/qupath/qupath/releases/latest
- Documentation : https://qupath.readthedocs.io/en/latest/
- Forum : https://forum.image.sc/tag/qupath



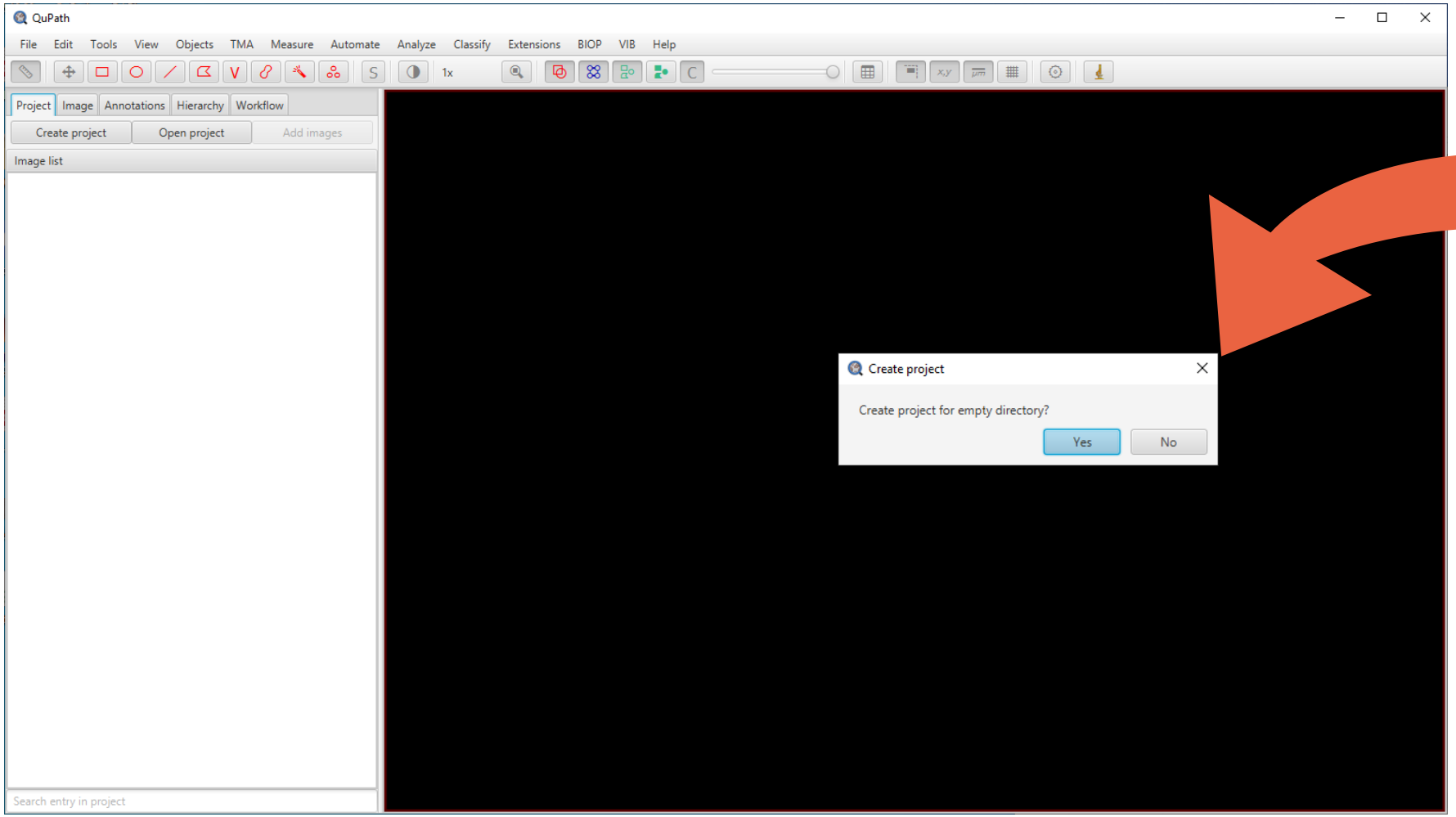
Peter Bankhead

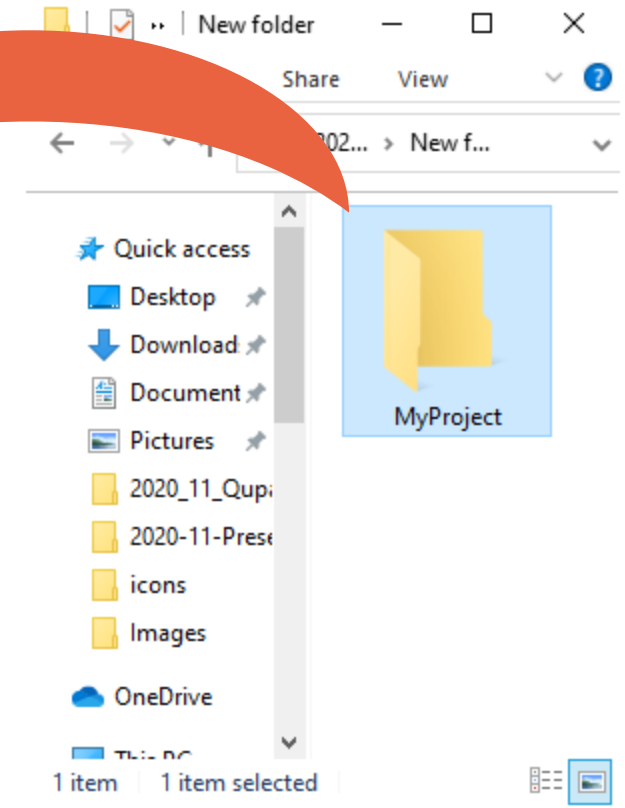
VIB

SCIENCE MEETS LIFE

QuPath : Create a Project

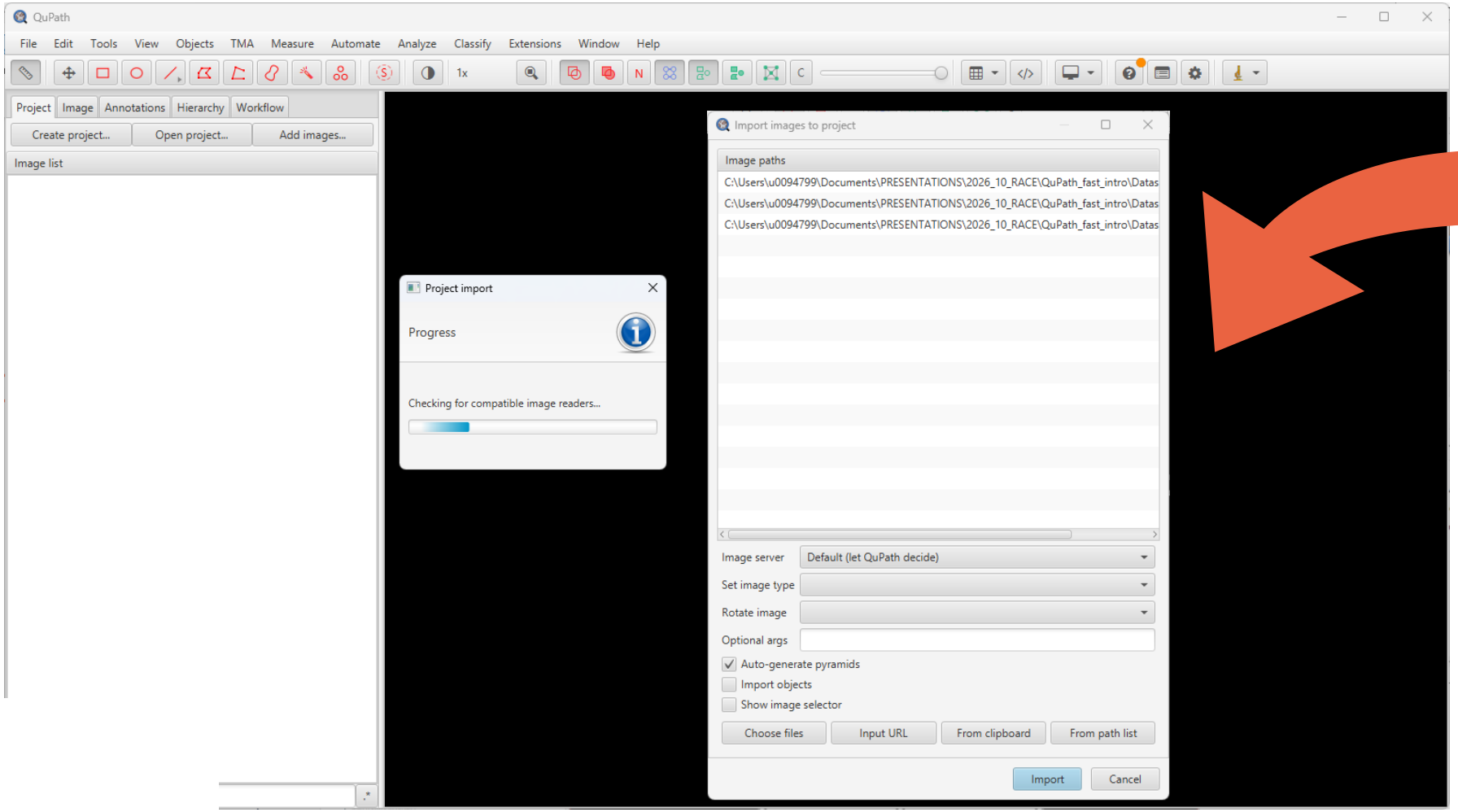
Drag a folder to create a project *or* File>Project>Create Project

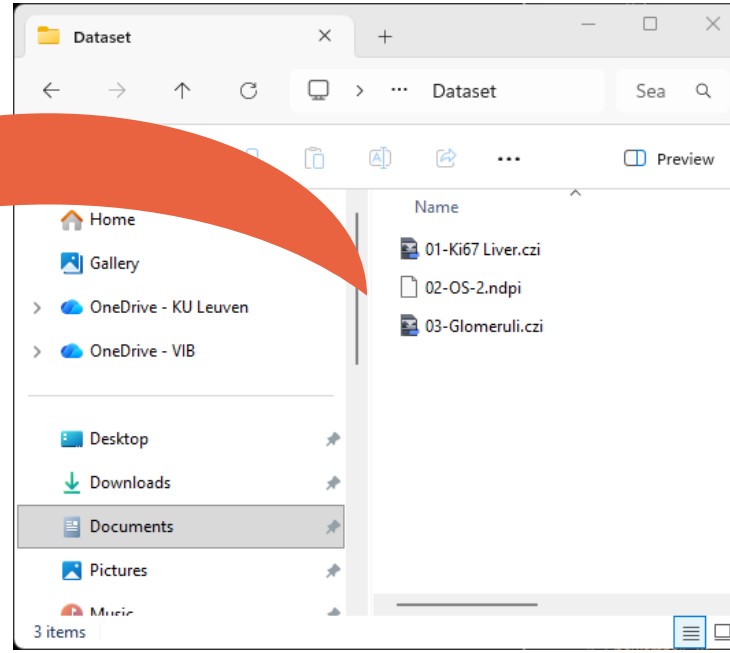




QuPath : Add images to project

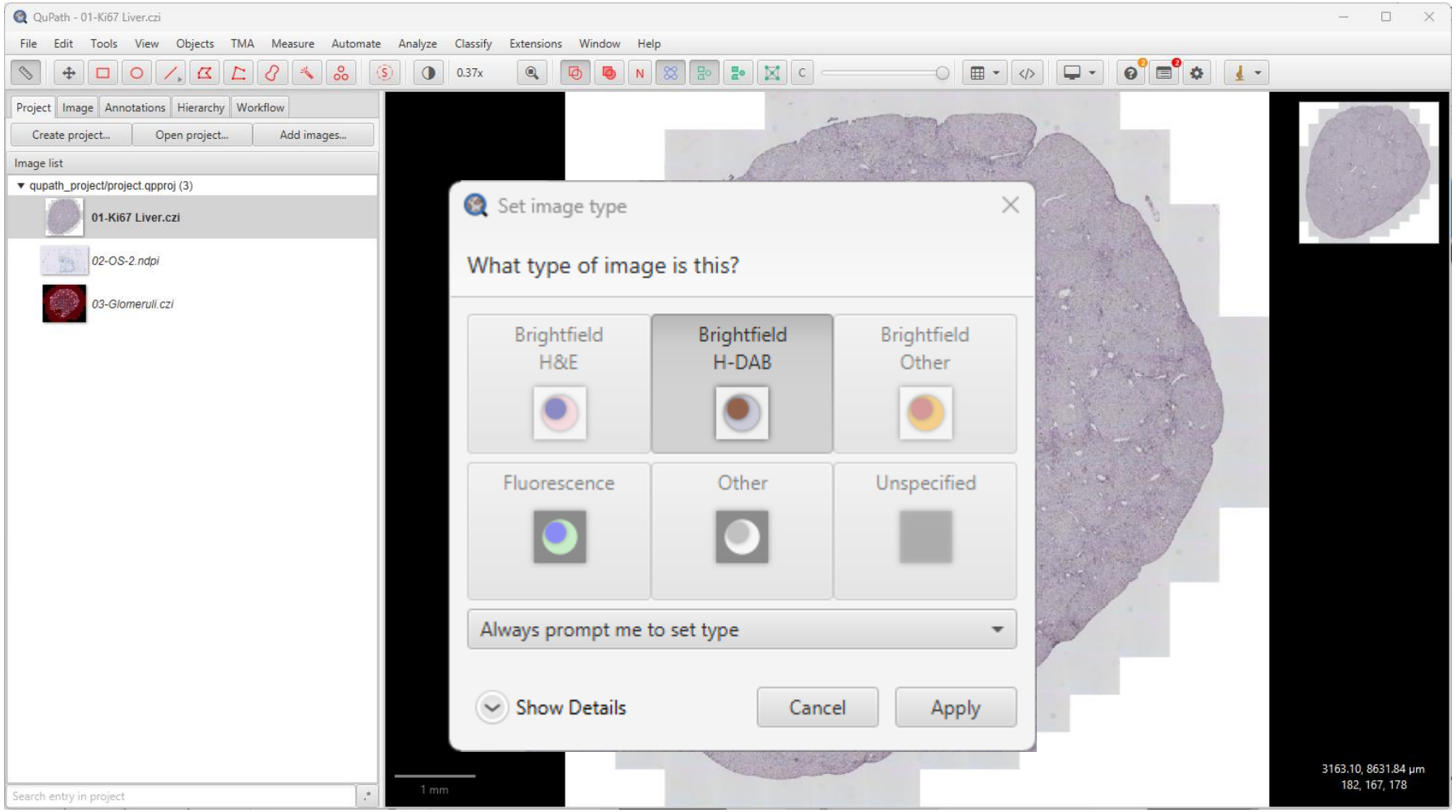
Drag and drop images you want to add





QuPath : Open an image

[Open Image 01-Ki67.czi]

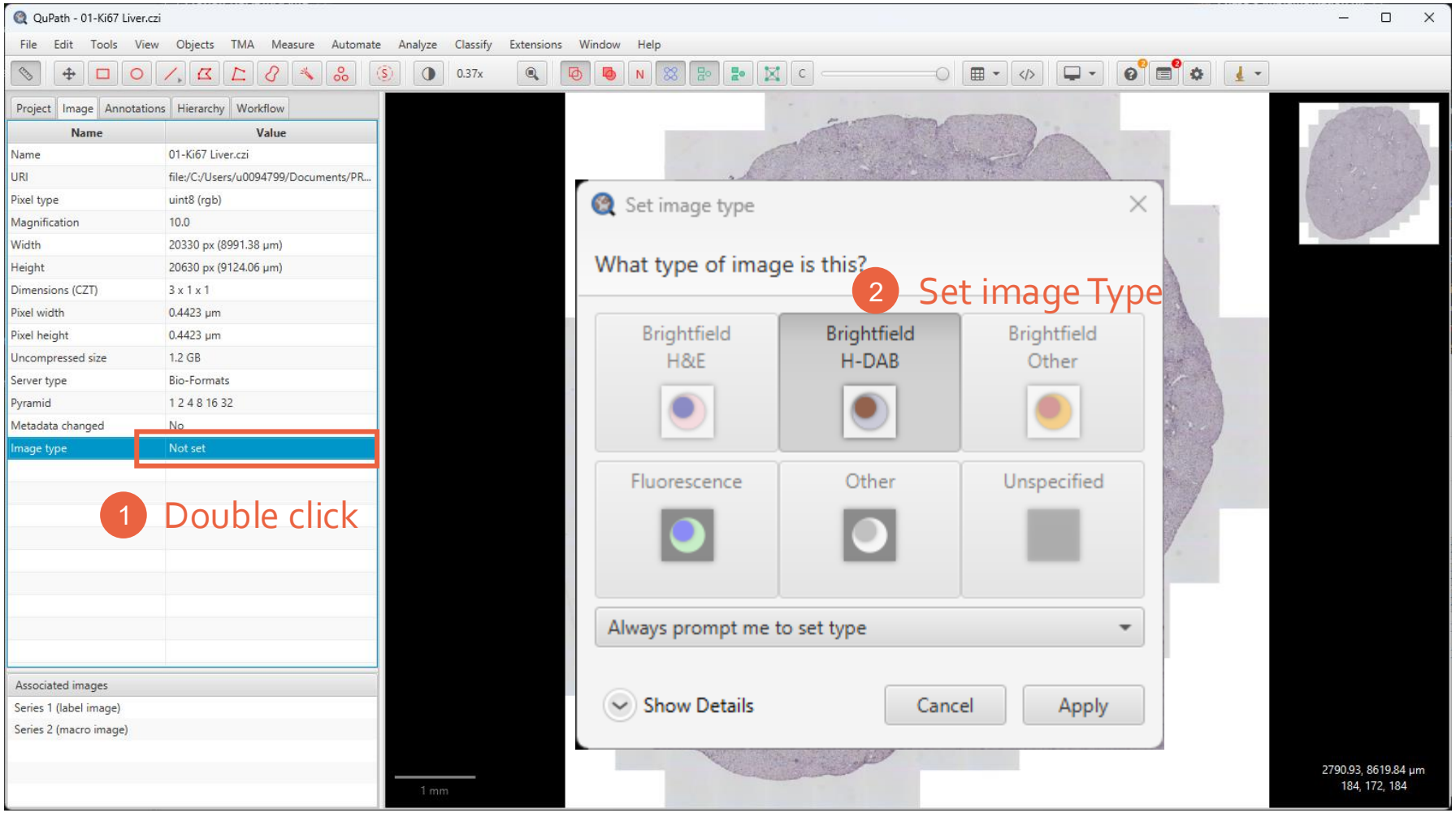


VIB

SCIENCE MEETS LIFE

QuPath : Change Image Type

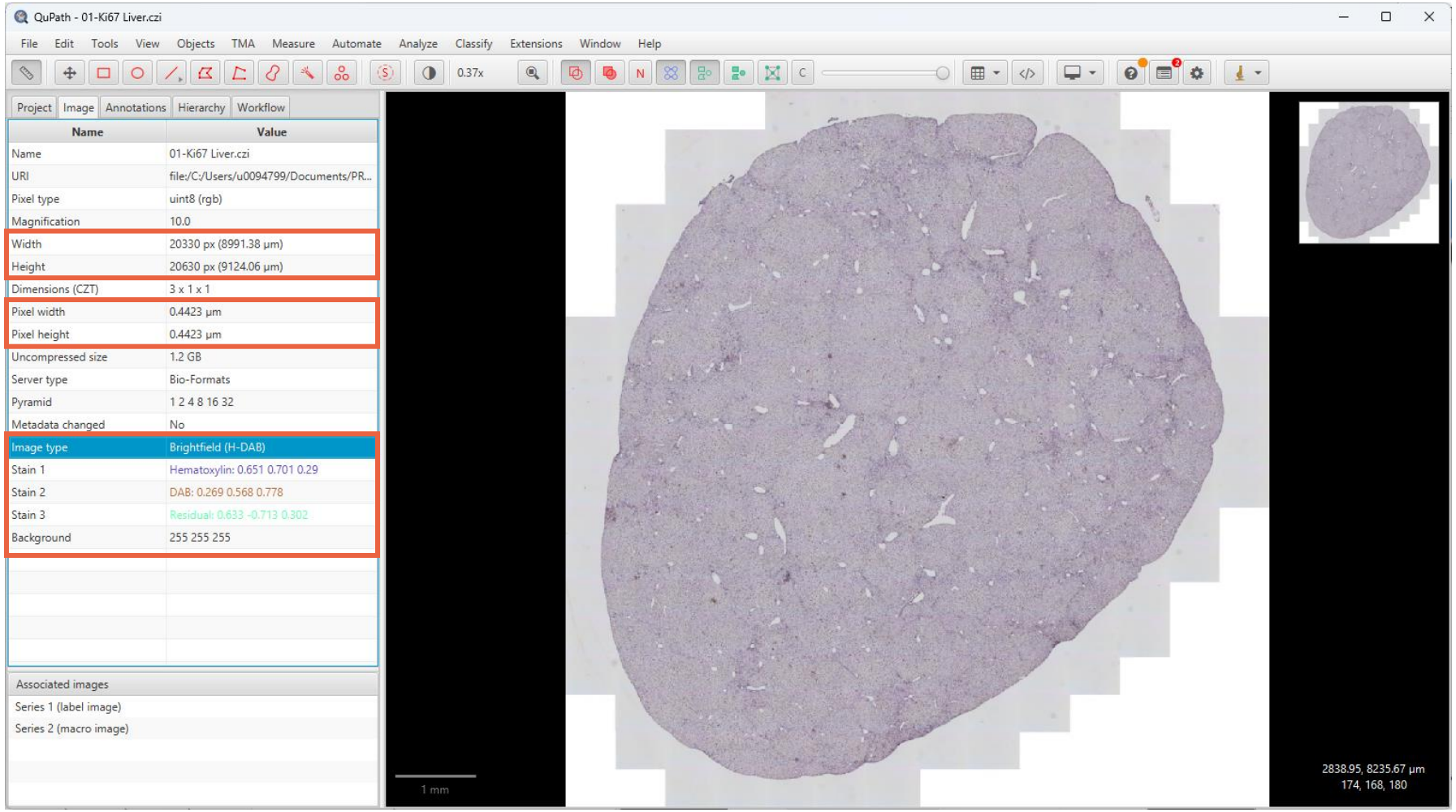
[Image Tab > Image Type]



SCIENCE MEETS LIFE

QuPath : Change Image Type

[Image Tab > Image Type]

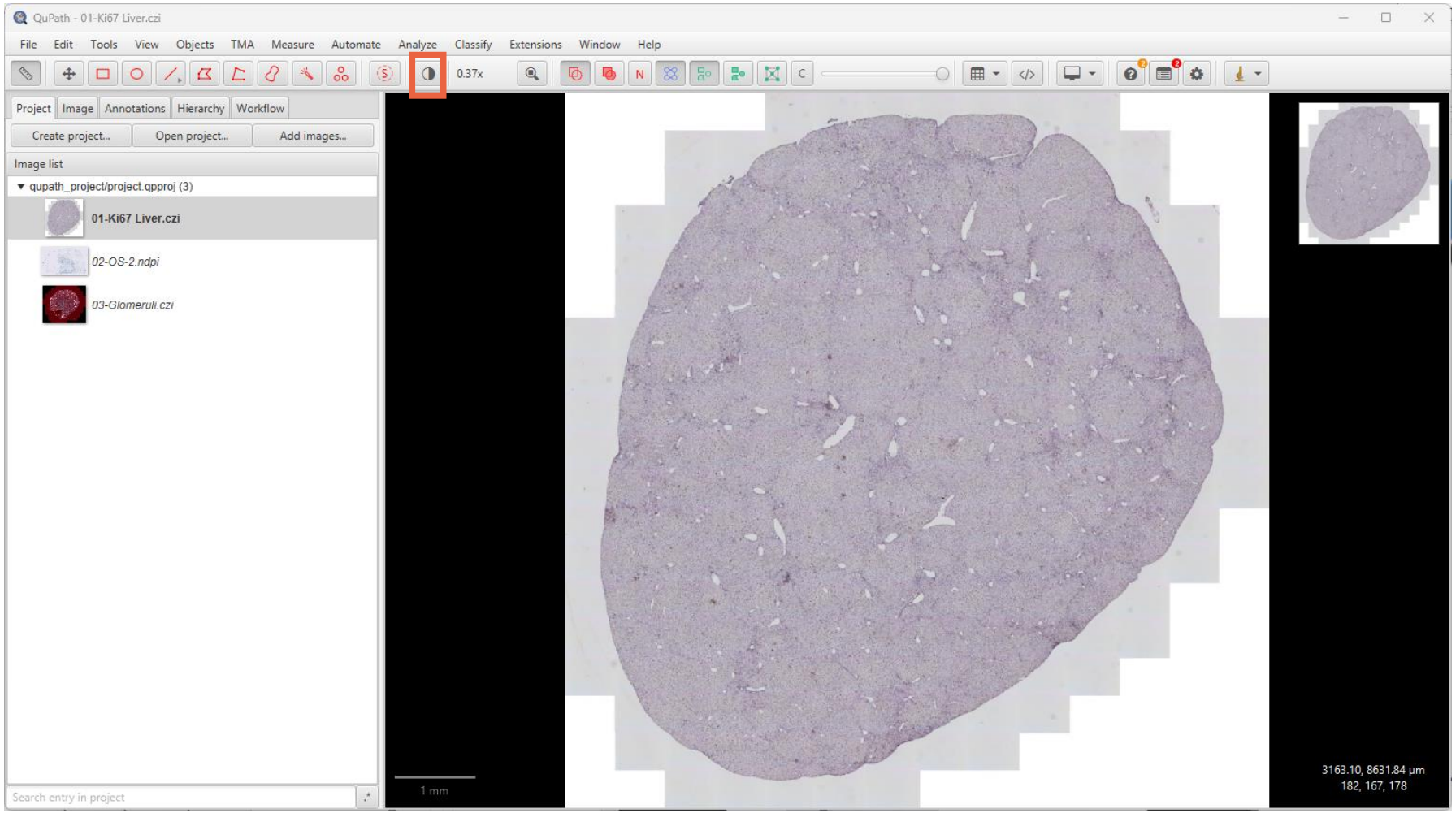


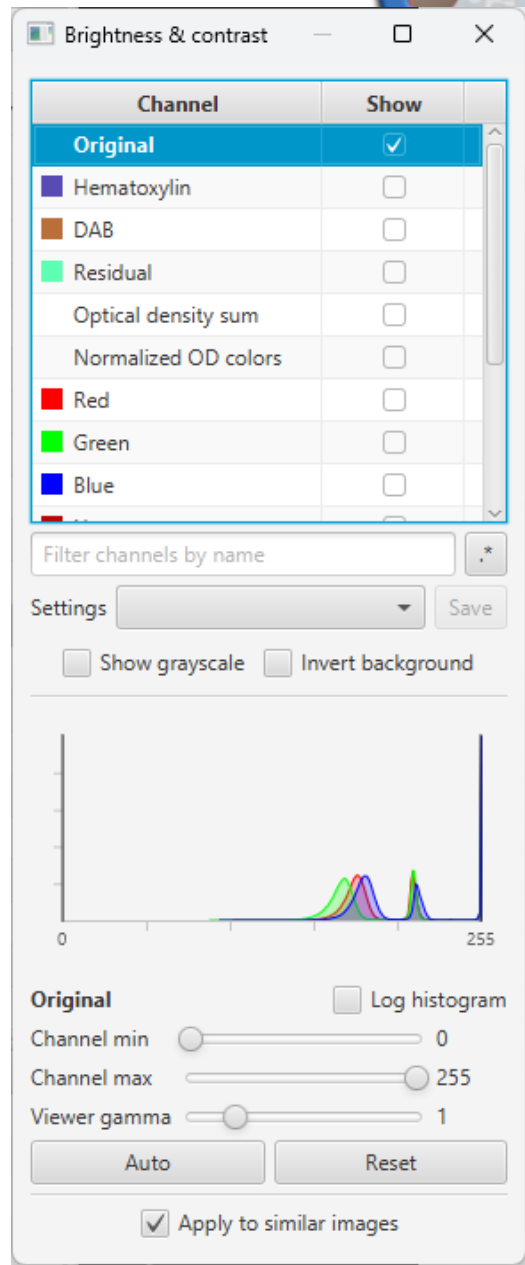
VIB

SCIENCE MEETS LIFE

QuPath : B&C, channels

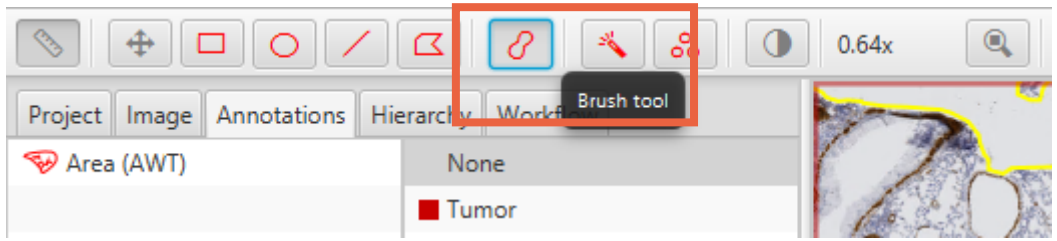
[View>Brightness/Contrast]





QuPath : Annotations

Use the Brush tool to modify the Area



Brush :

-add a new area, then select the 2 area and merge them **[Objects>Merge selected annotations]**

-remove a portion of the area**: alt+Brush (Option+Brush)** which will remove from the brush a portion of the selected ann

-avoid overlapping with another annotation using **ctrl+shift+Brush (Command+shift+Brush)**

Wand: to pick similar part of the image faster (same shortcut as the brush)

More info : https://qupath.readthedocs.io/en/latest/docs/starting/annotating.html

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QuPath : Annotations

Selection mode

One toolbar button that lives *beside* the annotation tools is not actually used to draw new annotations – but it is closely related.

The Selection mode button is used to temporarily repurpose the annotation tools so that they can be used to *select* objects (e.g. other annotations) by drawing around them.

If selection mode is turned on, using an annotation tool draws a shape with a dashed line to indicate that it isn't a regular annotation. This is indicated by the button changing to a dashed version of the icon.



Selection version of the drawing tools

Instead, it's defining a region, and objects inside that region will be selected when the drawing is finished.

More info : https://qupath.readthedocs.io/en/latest/docs/starting/annotating.html

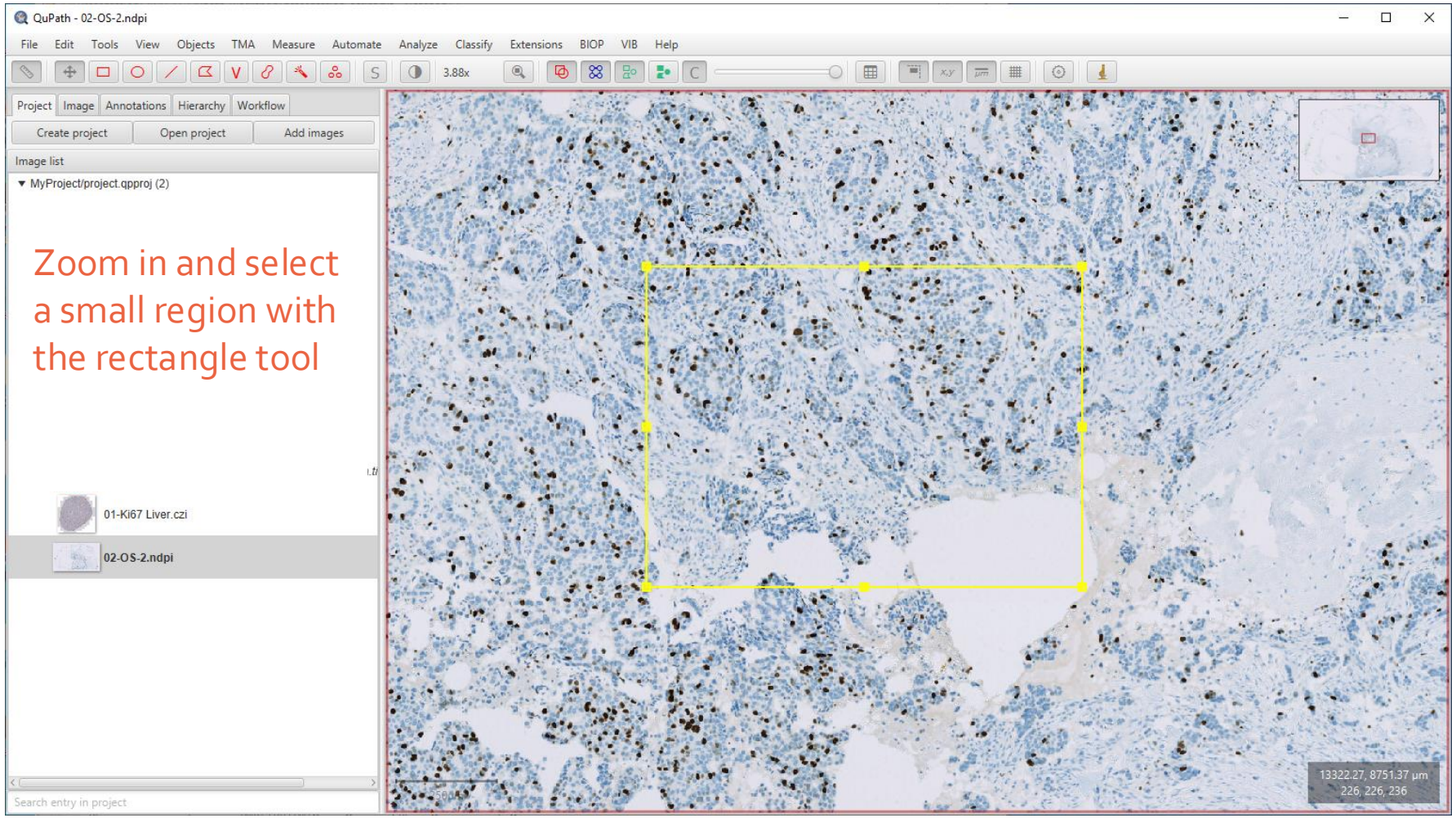
SCIENCE MEETS LIFE

QuPath : Stain Editor

1-Draw a rectangle



02-OS-2.ndpi



VIB

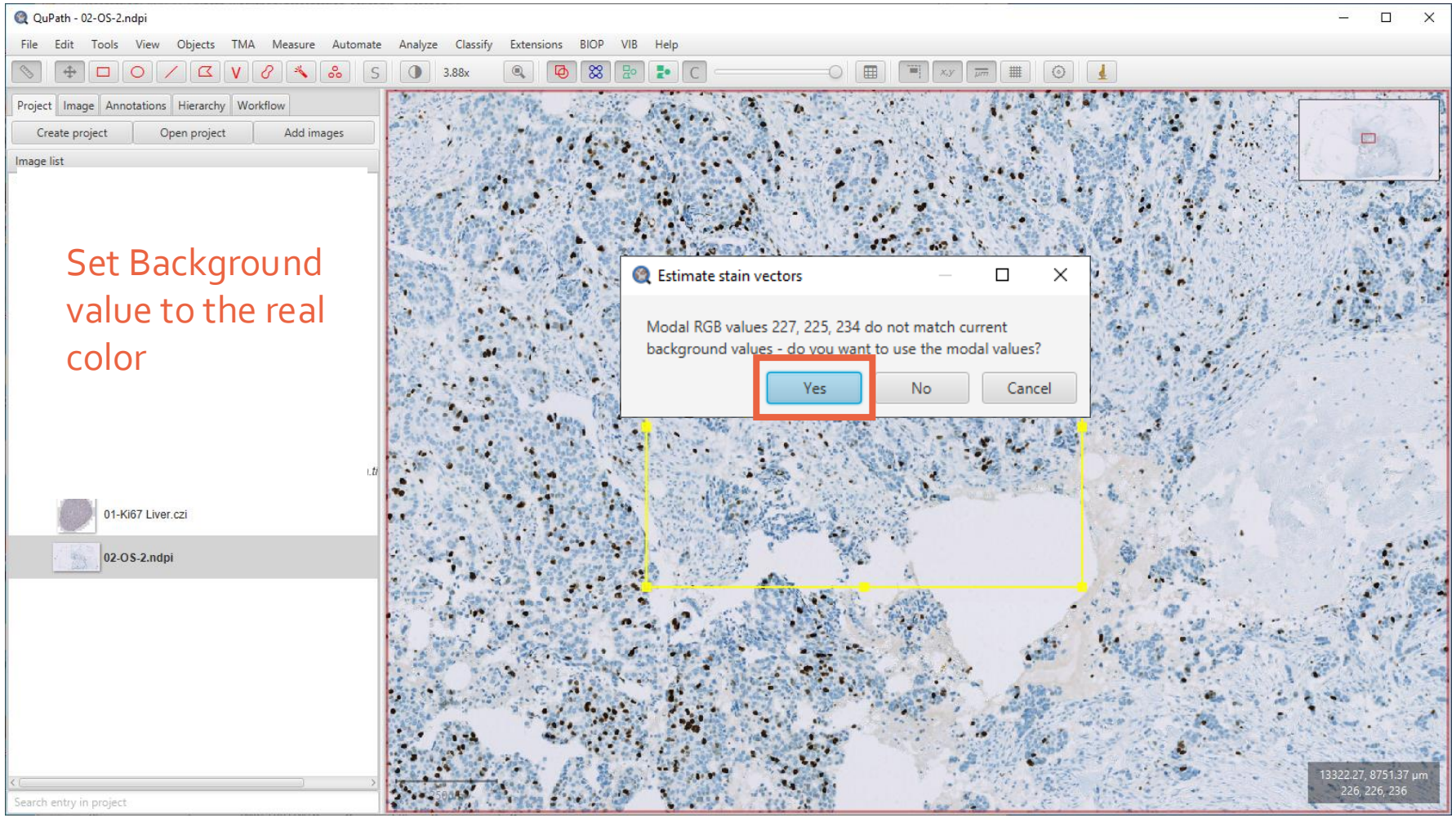
SCIENCE MEETS LIFE

QuPath : Stain Editor

2- Analyze>Estimate Stain Vector



02-OS-2.ndpi



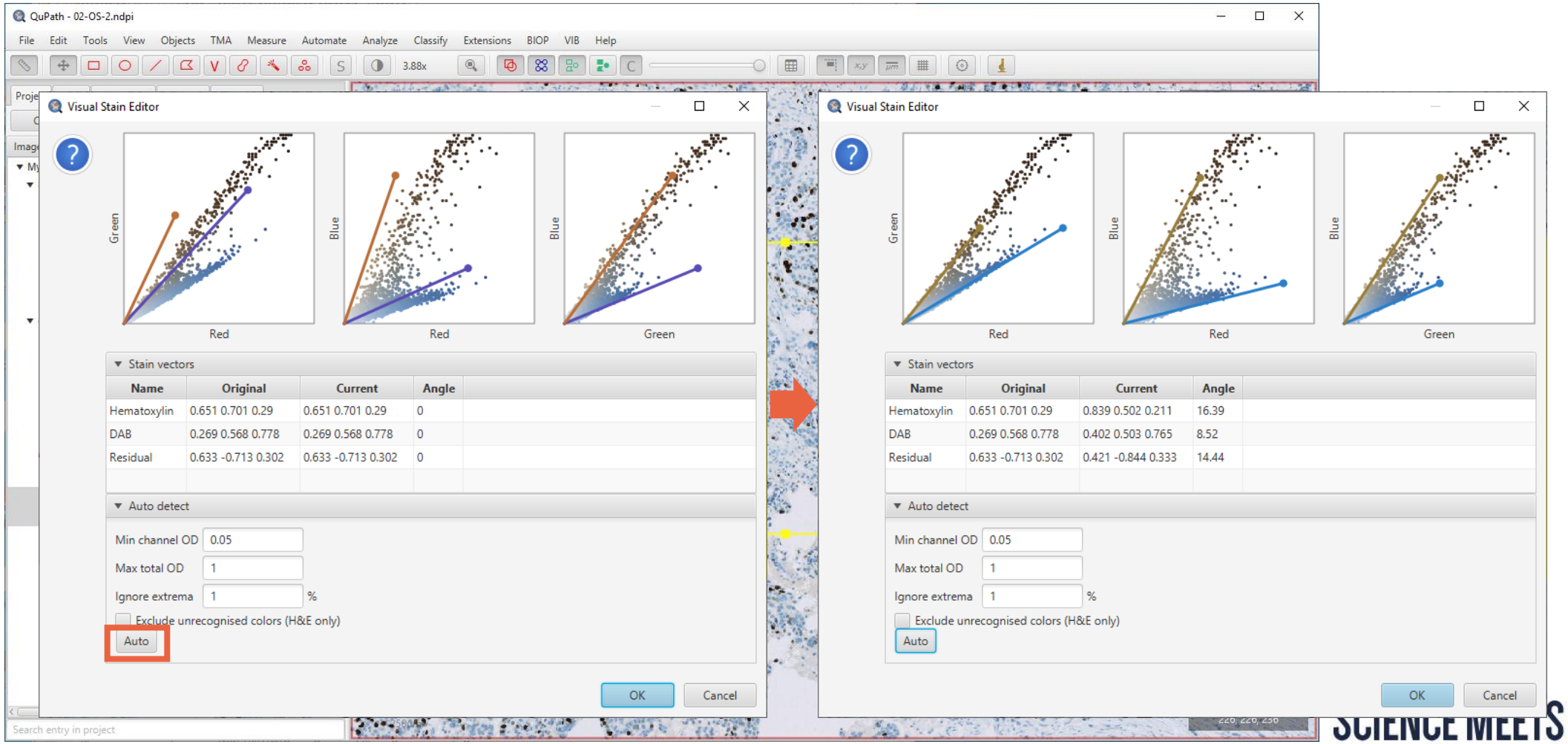
VIB

SCIENCE MEETS LIFE

QuPath : Stain Editor

Transform the stain vector with auto estimation

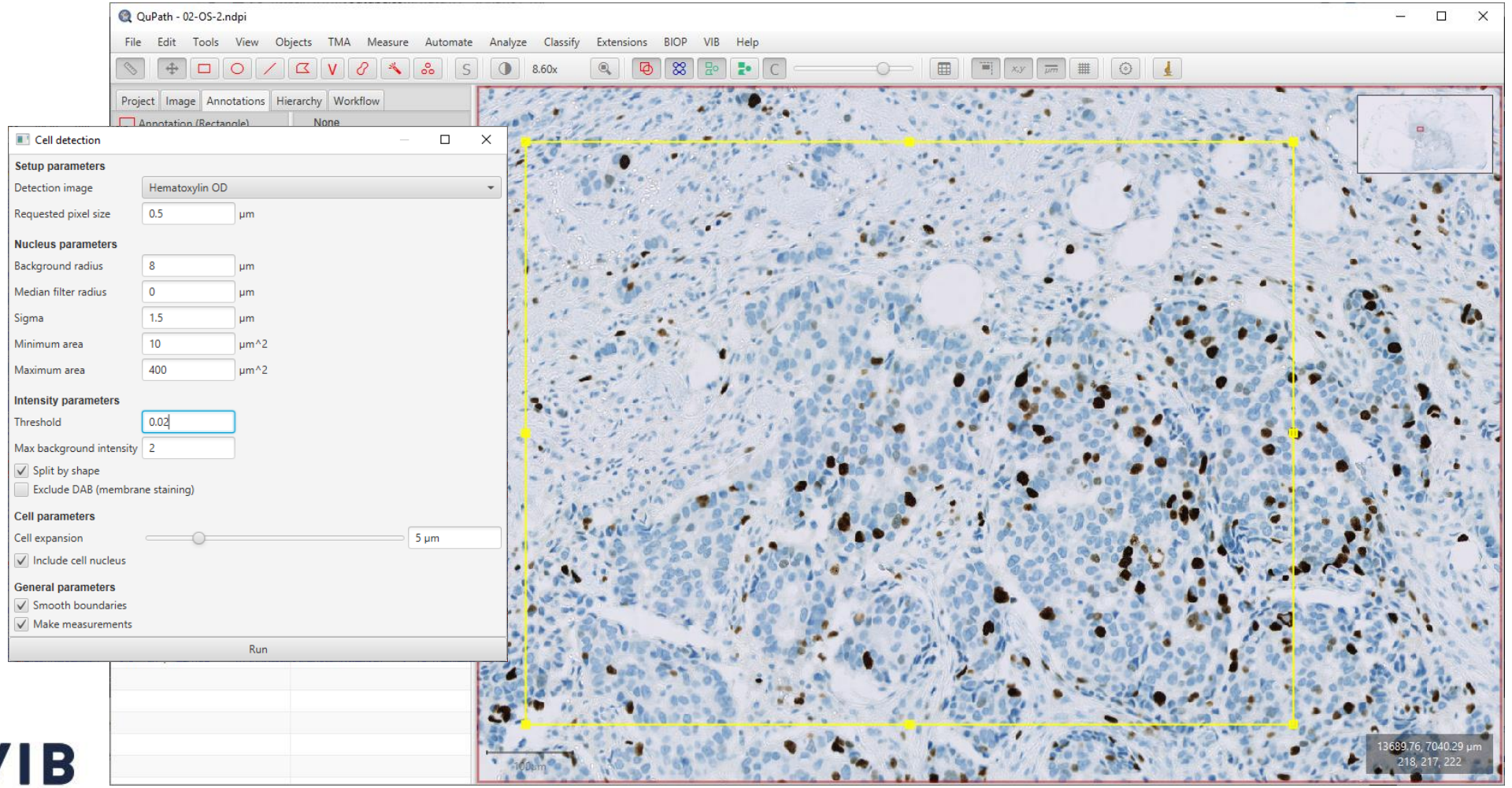
02-OS-2.ndpi



QuPath : Cell Detection

[Analyze> Cell detection > Cell detection]

02-OS-2.ndpi



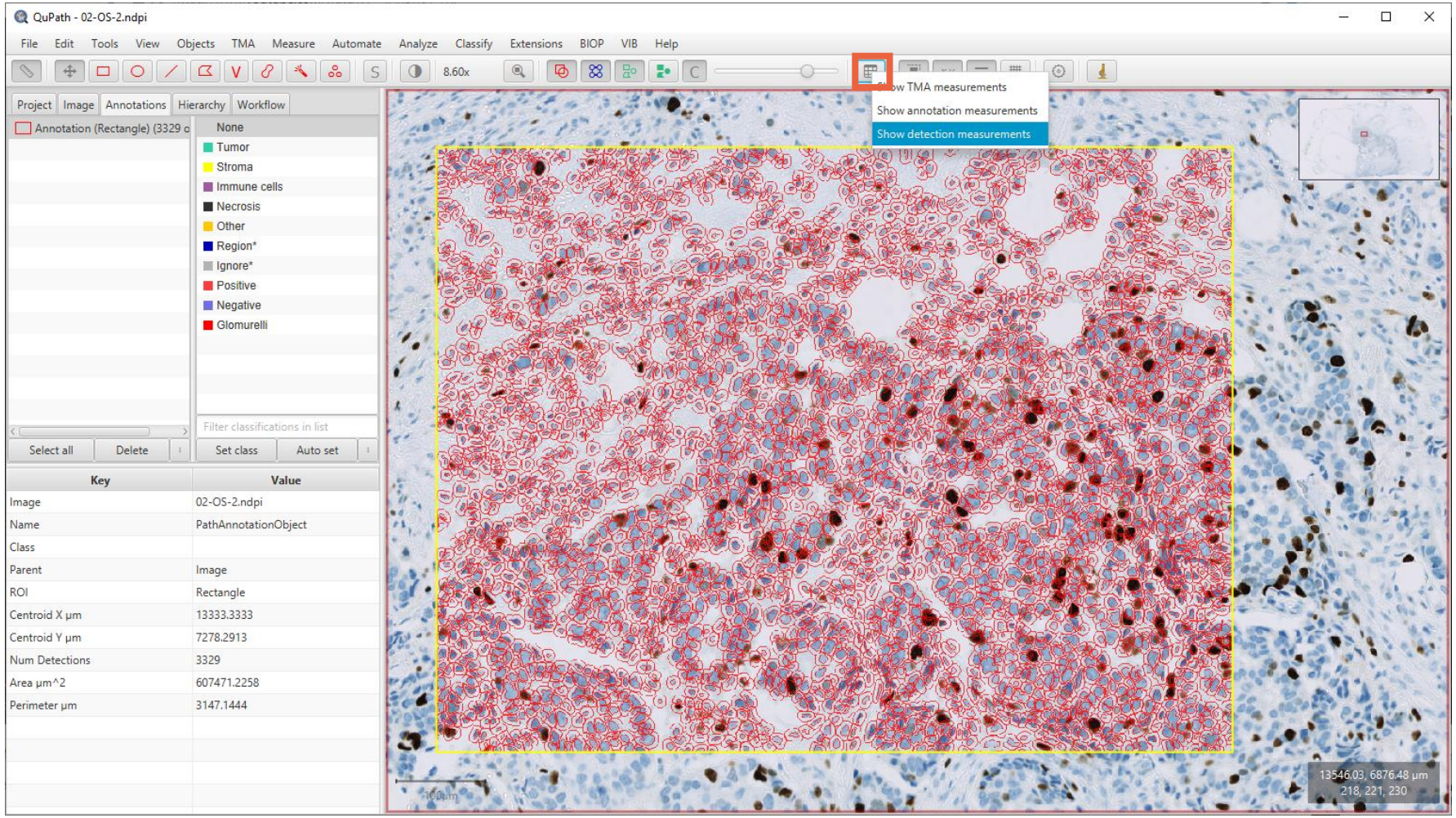
VIB SCIENCE MEETS LIFE

QuPath : Cell Detection

[Measure> Show detection measurements]



02-OS-2.ndpi



Cell are measured automatically
View the measurements per cell (detection)

VIB

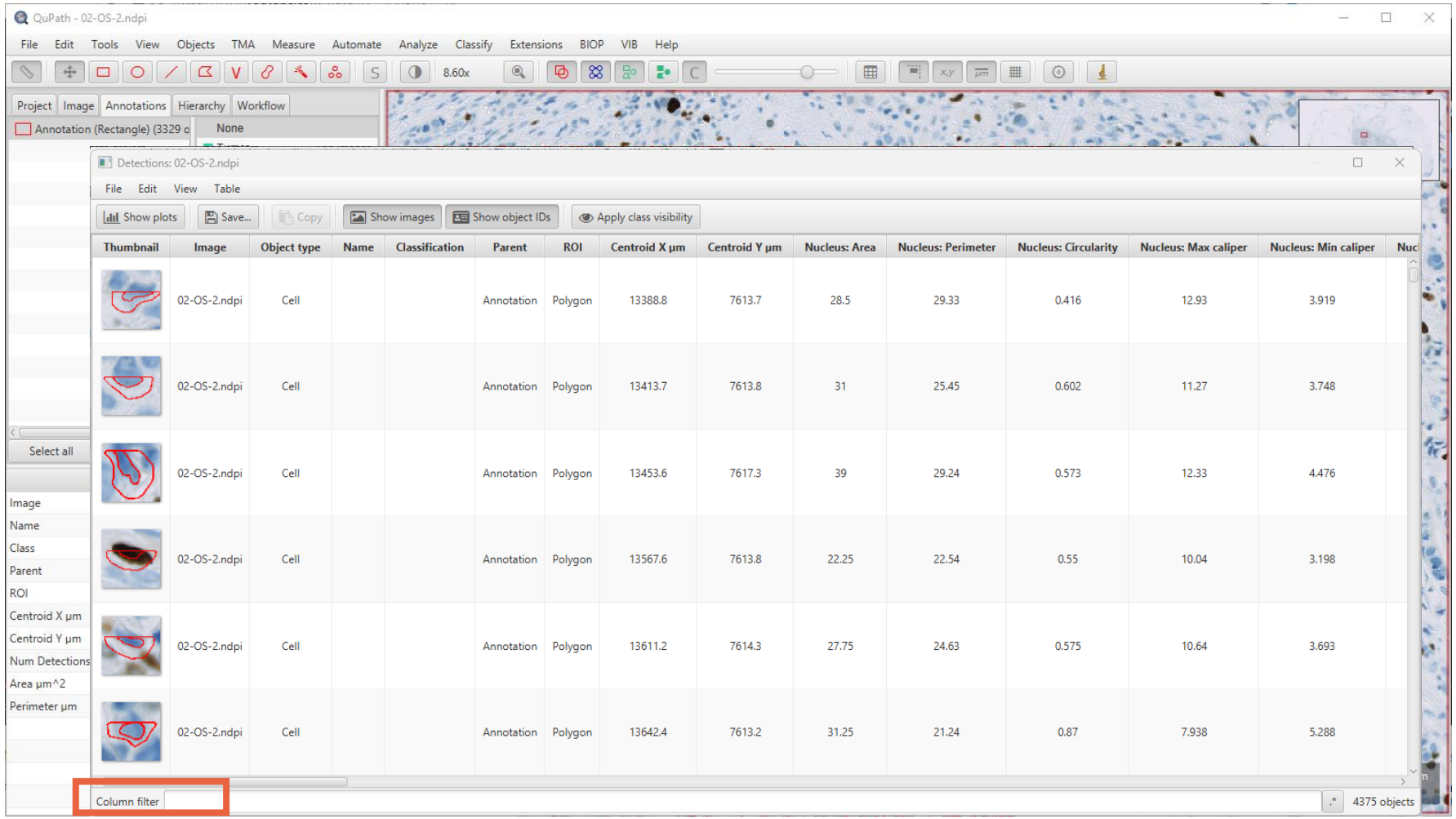
SCIENCE MEETS LIFE

QuPath : Cell Detection

[Measure> Show detection measurements]



02-OS-2.ndpi



Select a row in the table to see the cell

Filter by column

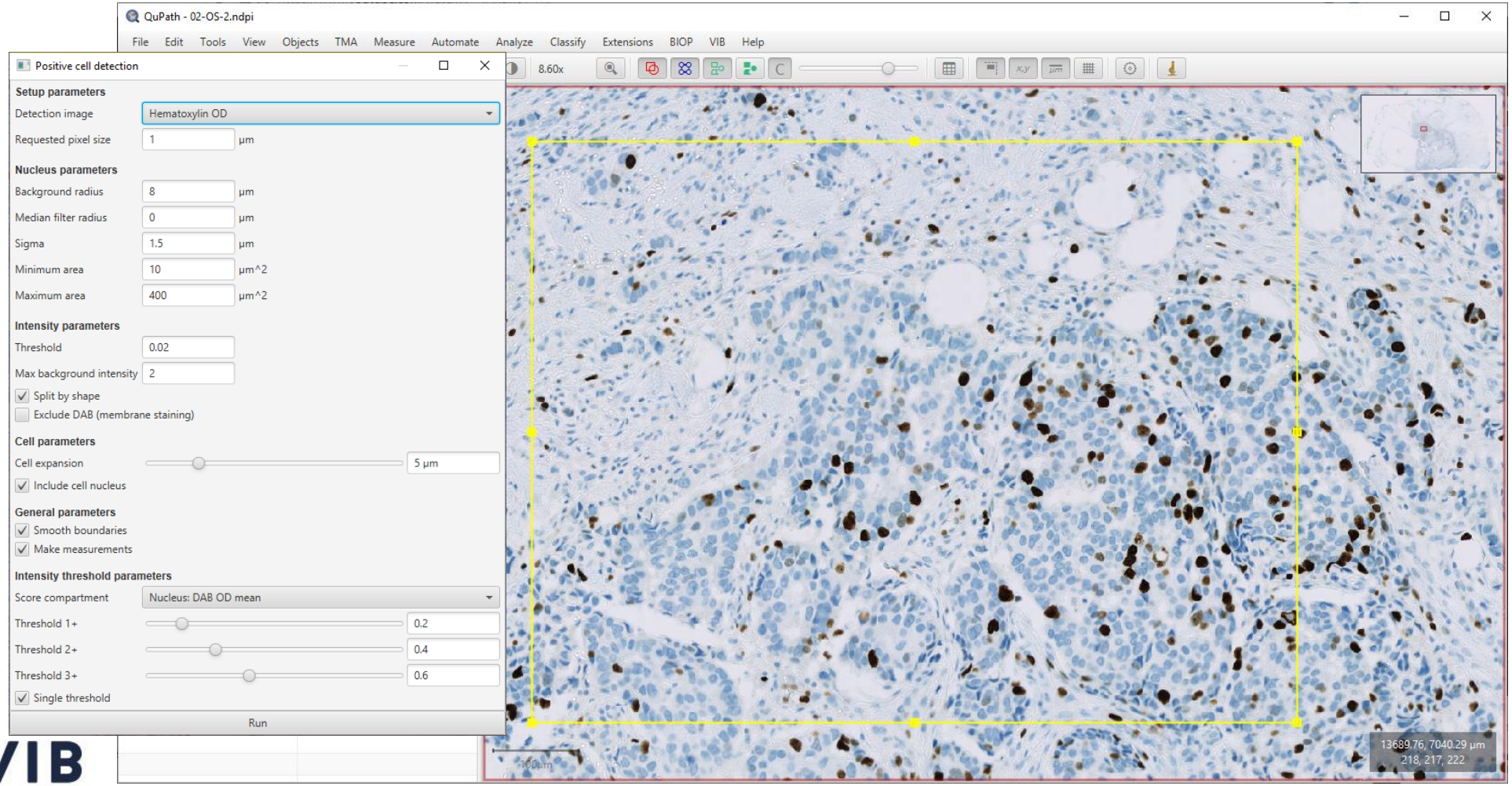
VIB

SCIENCE MEETS LIFE

QuPath : Cell Detection

[Analyze> Cell detection > Positive cell detection]

02-OS-2.ndpi



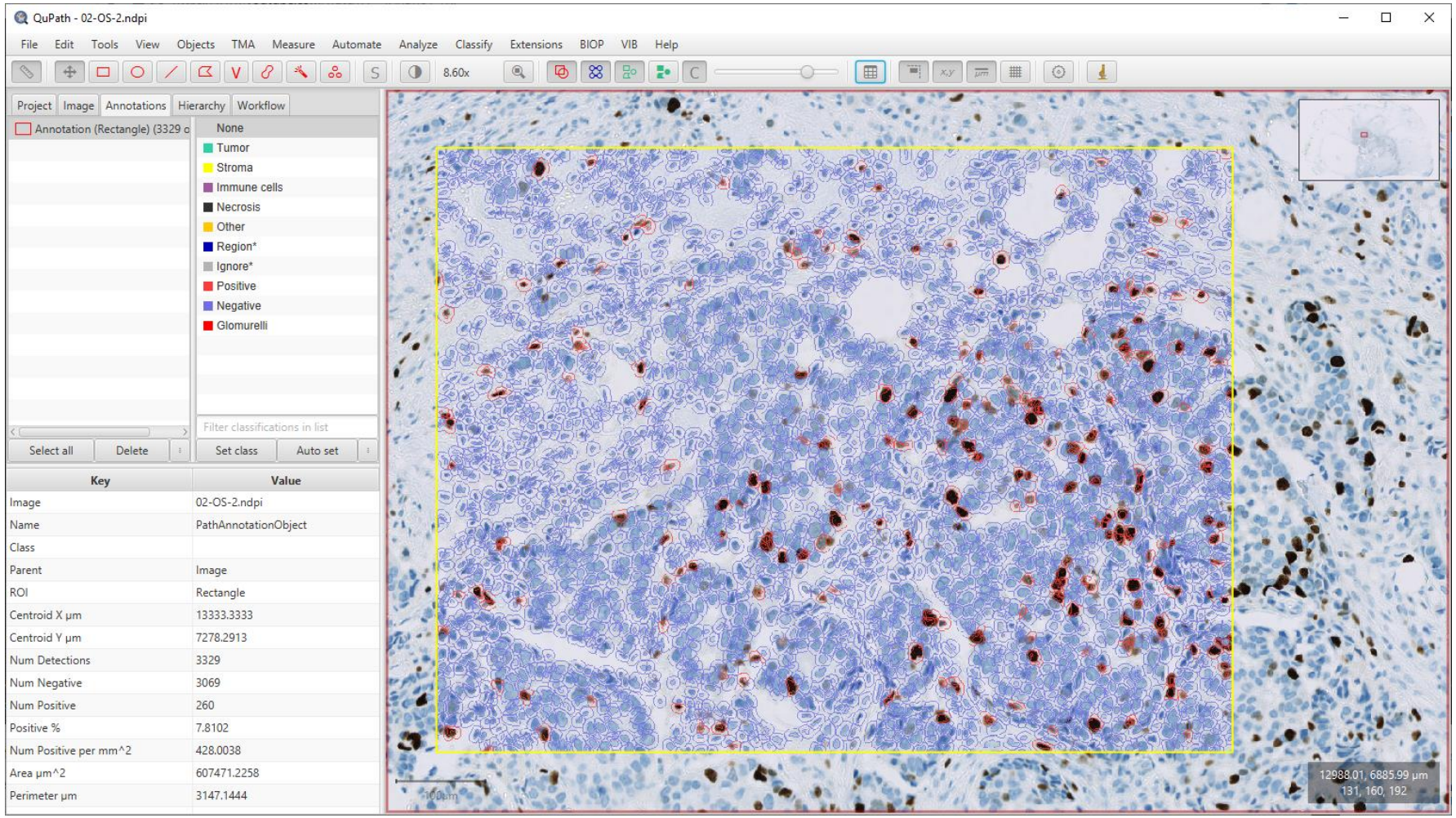
VIB SCIENCE MEETS LIFE

QuPath : Cell Detection

[Analyze> Cell detection > Positive cell detection]



02-OS-2.ndpi





Positive cell



Negative cell

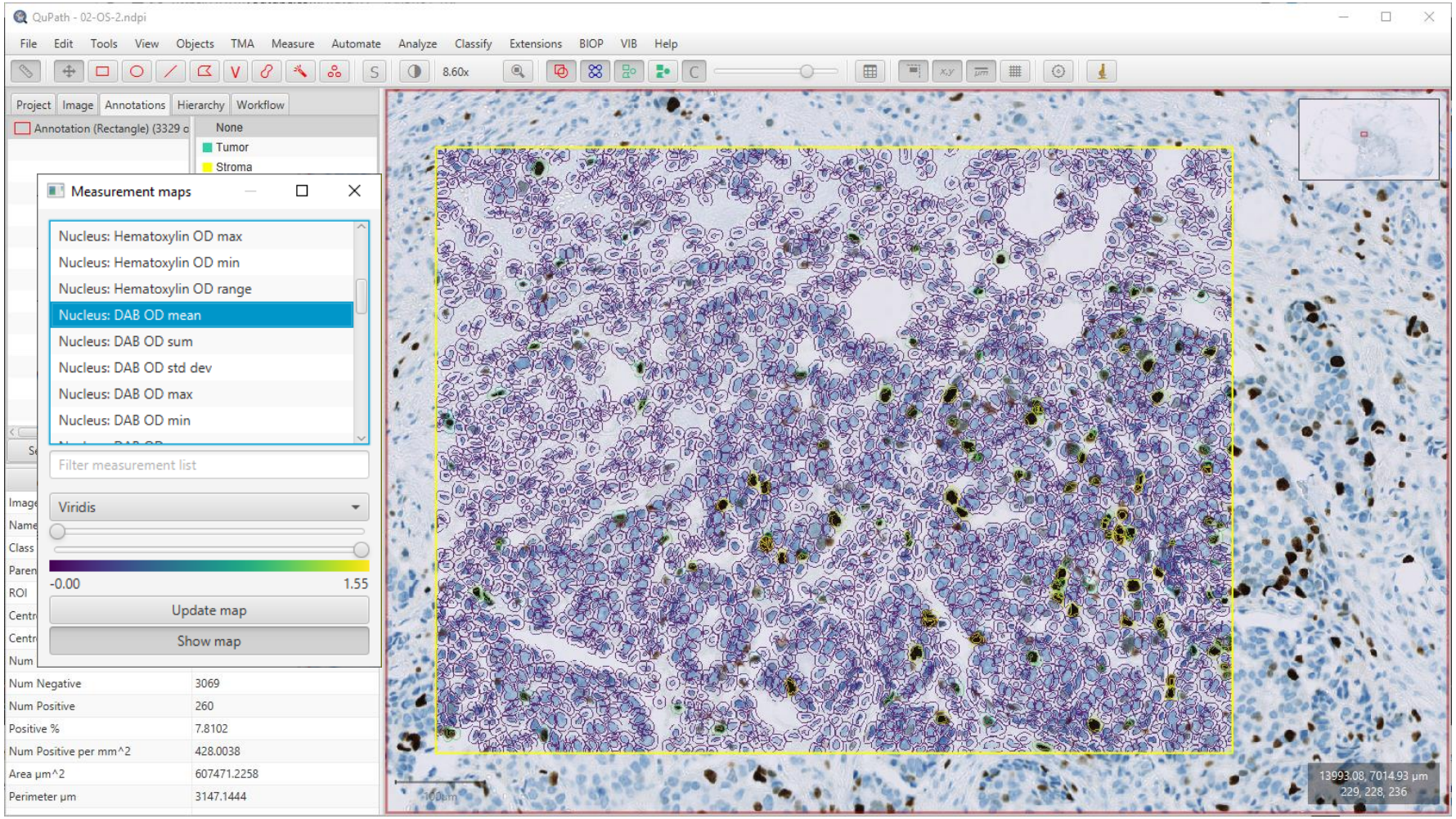
VIB

SCIENCE MEETS LIFE

QuPath : Cell Detection

[Measure> Show measurement map : Nucleus DAB OD mean]

02-OS-2.ndpi



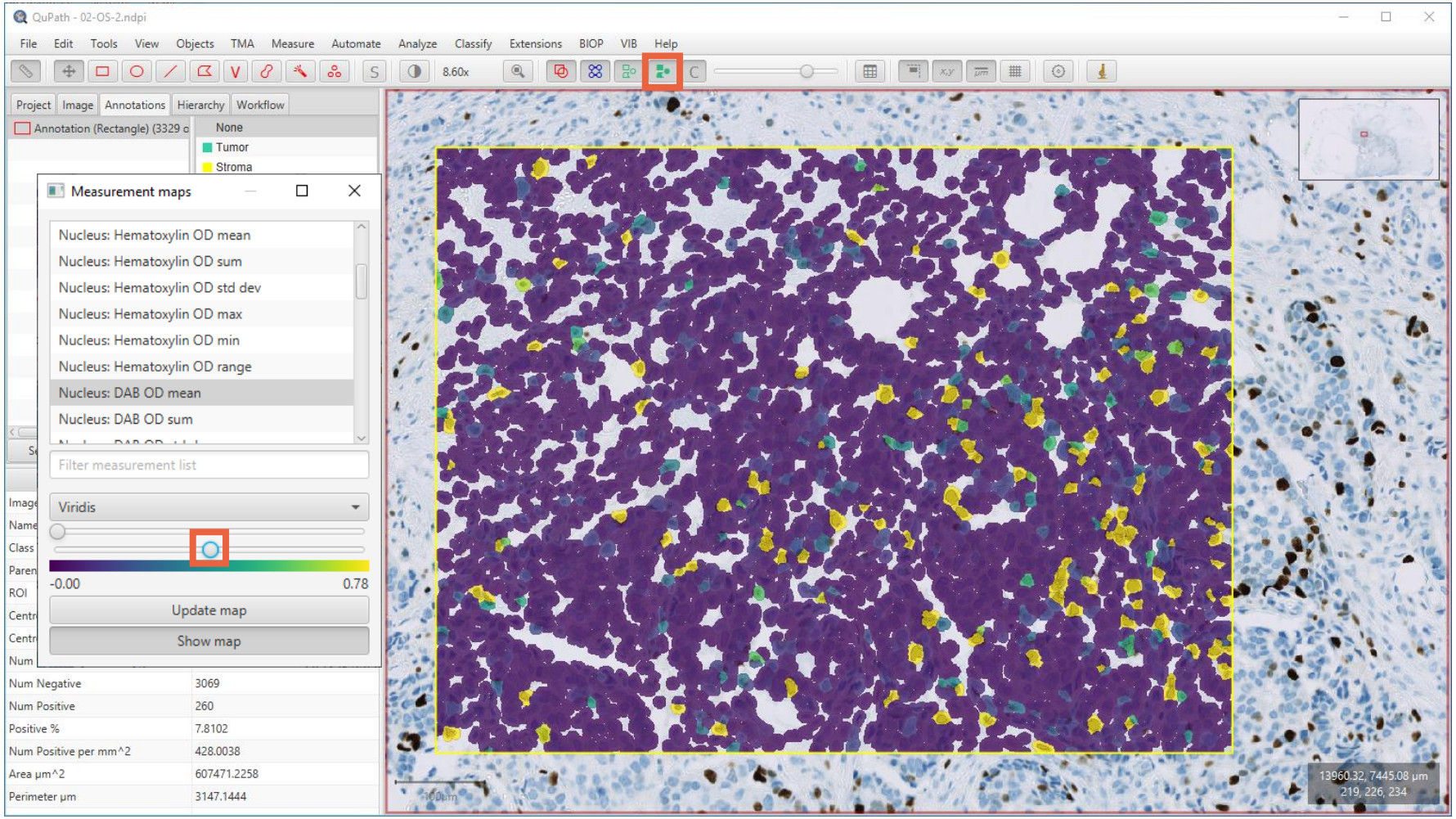
VIB

SCIENCE MEETS LIFE

QuPath : Cell Detection

[View> Fill detection]

02-OS-2.ndpi



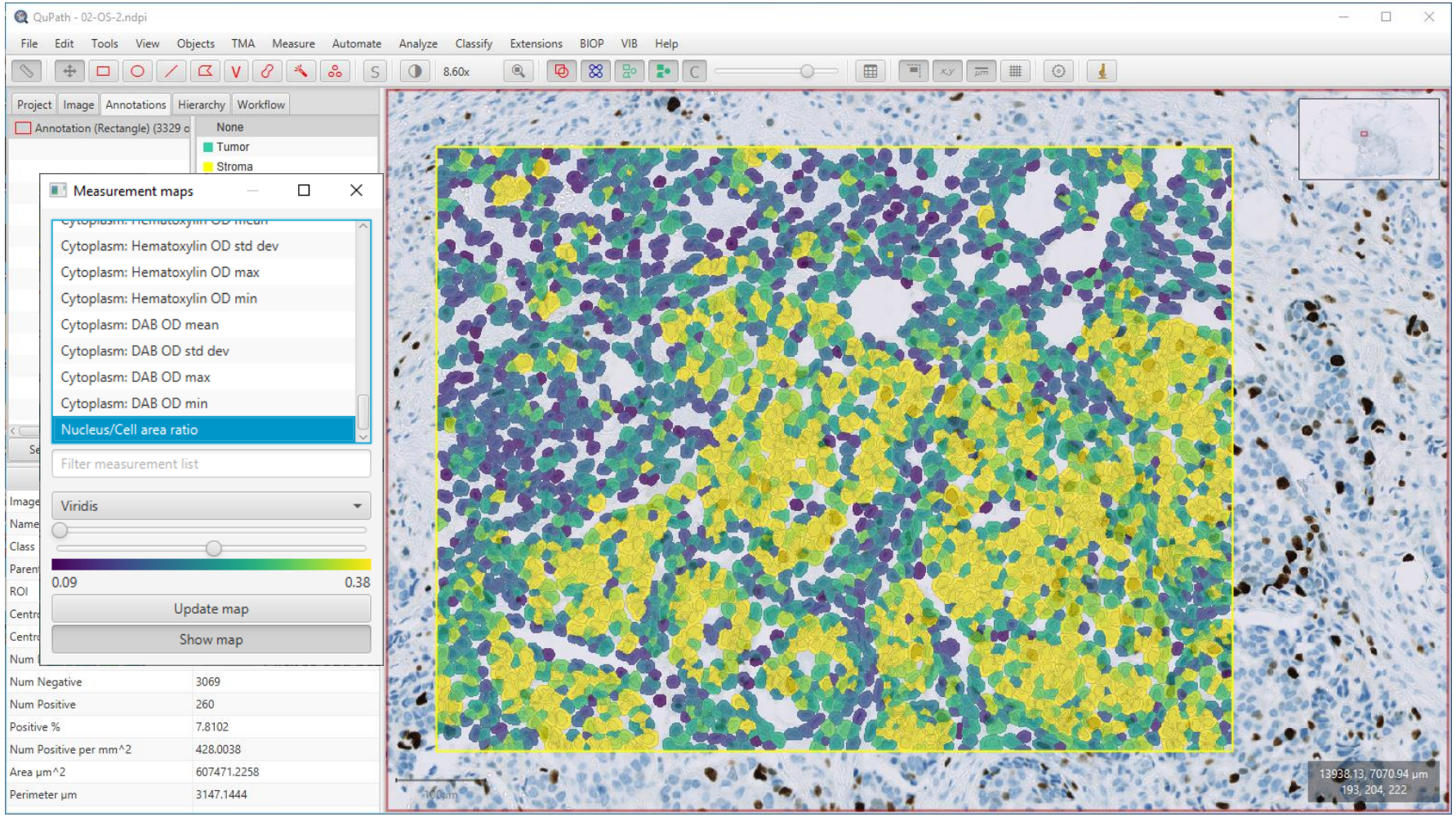
VIB

SCIENCE MEETS LIFE

QuPath : Cell Detection

[Measure> Show measurement map : Nucleus/Cell area ratio]

02-OS-2.ndpi



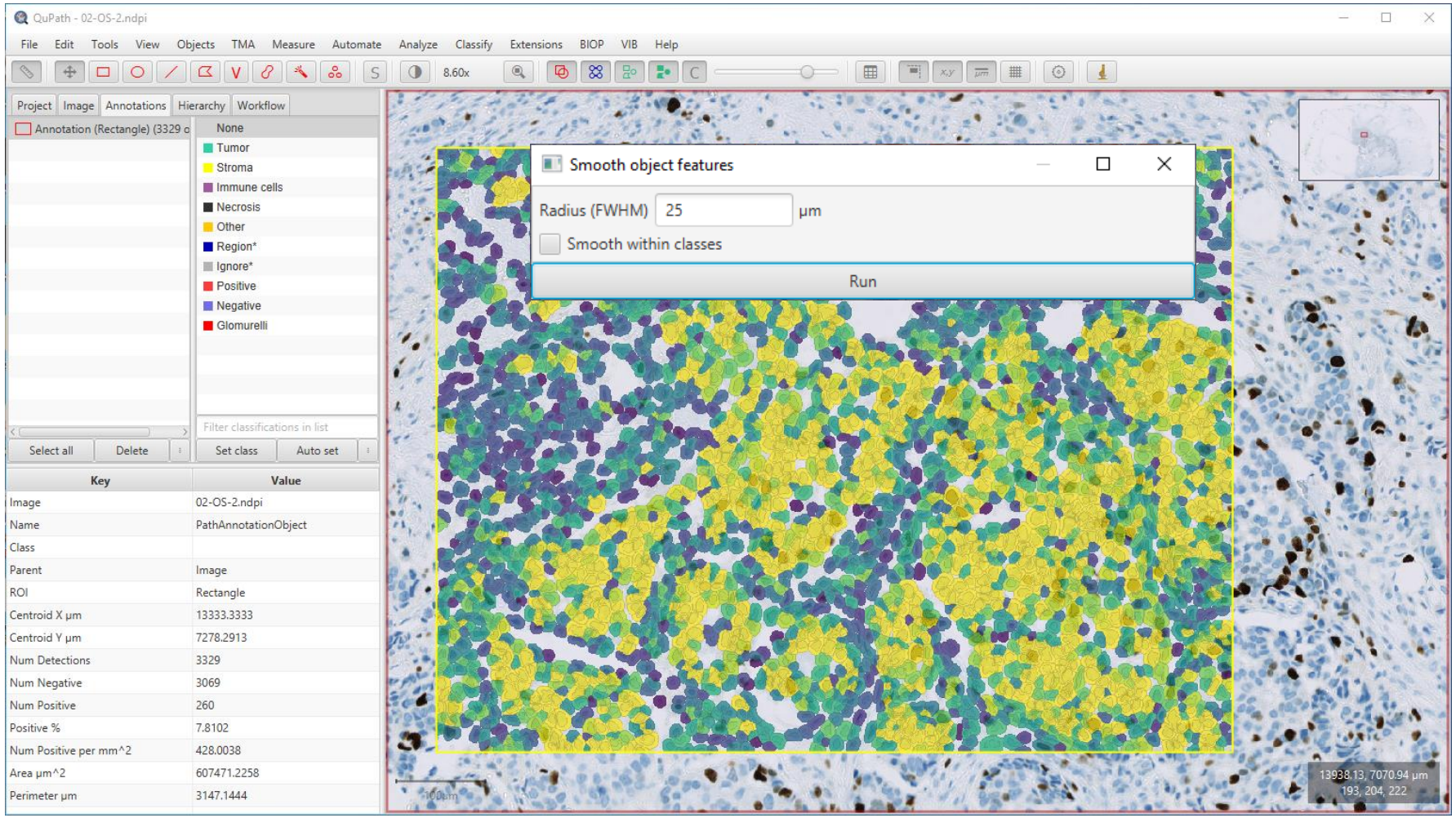
VIB

SCIENCE MEETS LIFE

QuPath : Cell Detection

[Analyze>Calculate Features>Add smooth features]

02-OS-2.ndpi



Weight the measurement with the neighboured cells

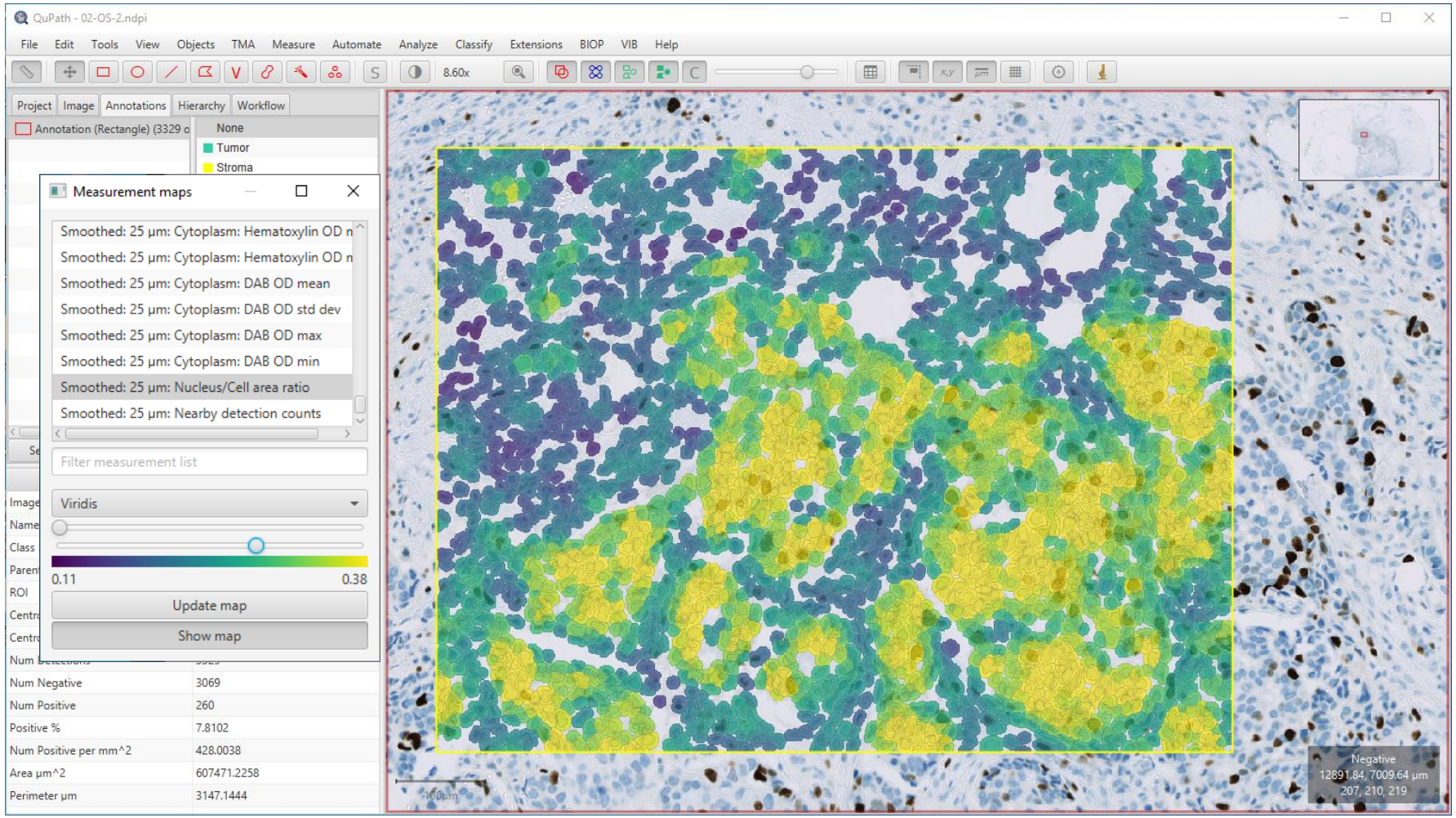
VIB

SCIENCE MEETS LIFE

QuPath : Cell Detection

[Measure> Show measurement map : Nucleus/Cell area ratio]

02-OS-2.ndpi



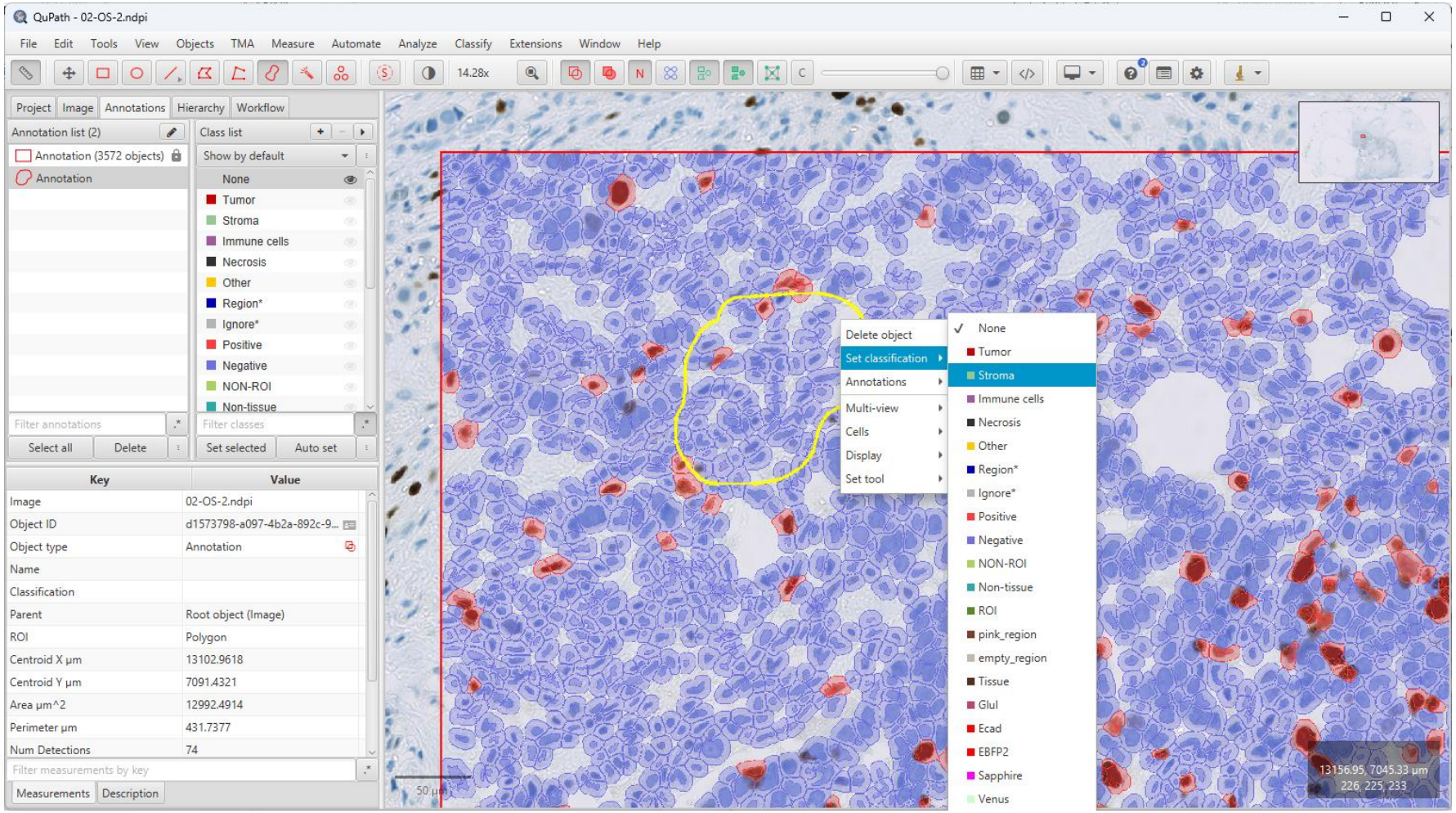
VIB

SCIENCE MEETS LIFE

QuPath : Train object classifier

[Create 2 annotations, tumor and stroma and assigned classes]

02-OS-2.ndpi



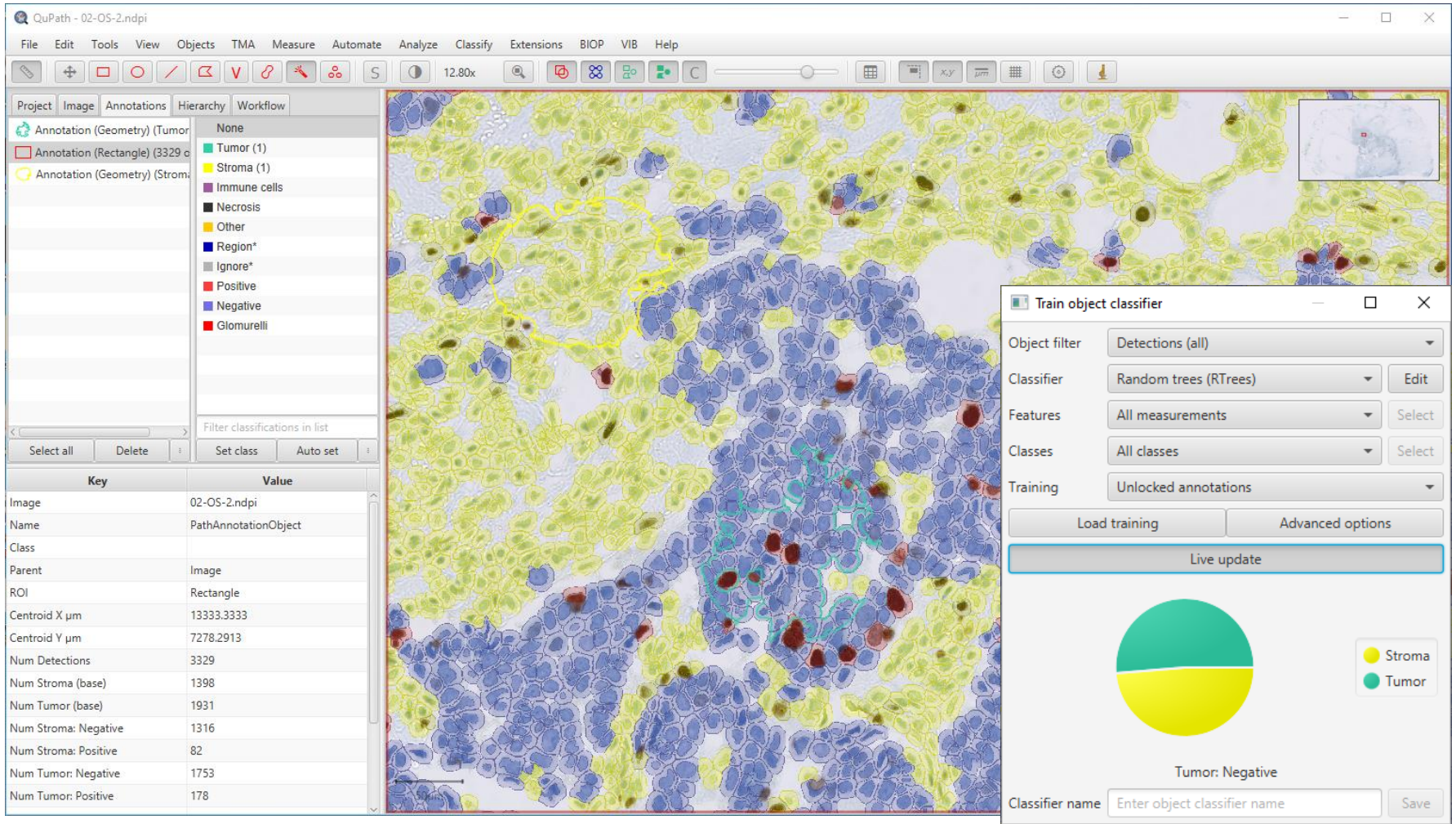
SCIENCE MEETS LIFE

QuPath : Train object classifier

[Classify>Object classification>Train Object classifier]



02-OS-2.ndpi



Separate in stroma area and tumor area.

Now, **red** is tumor cells positive, **blue** is tumor negative and **yellow** is stroma

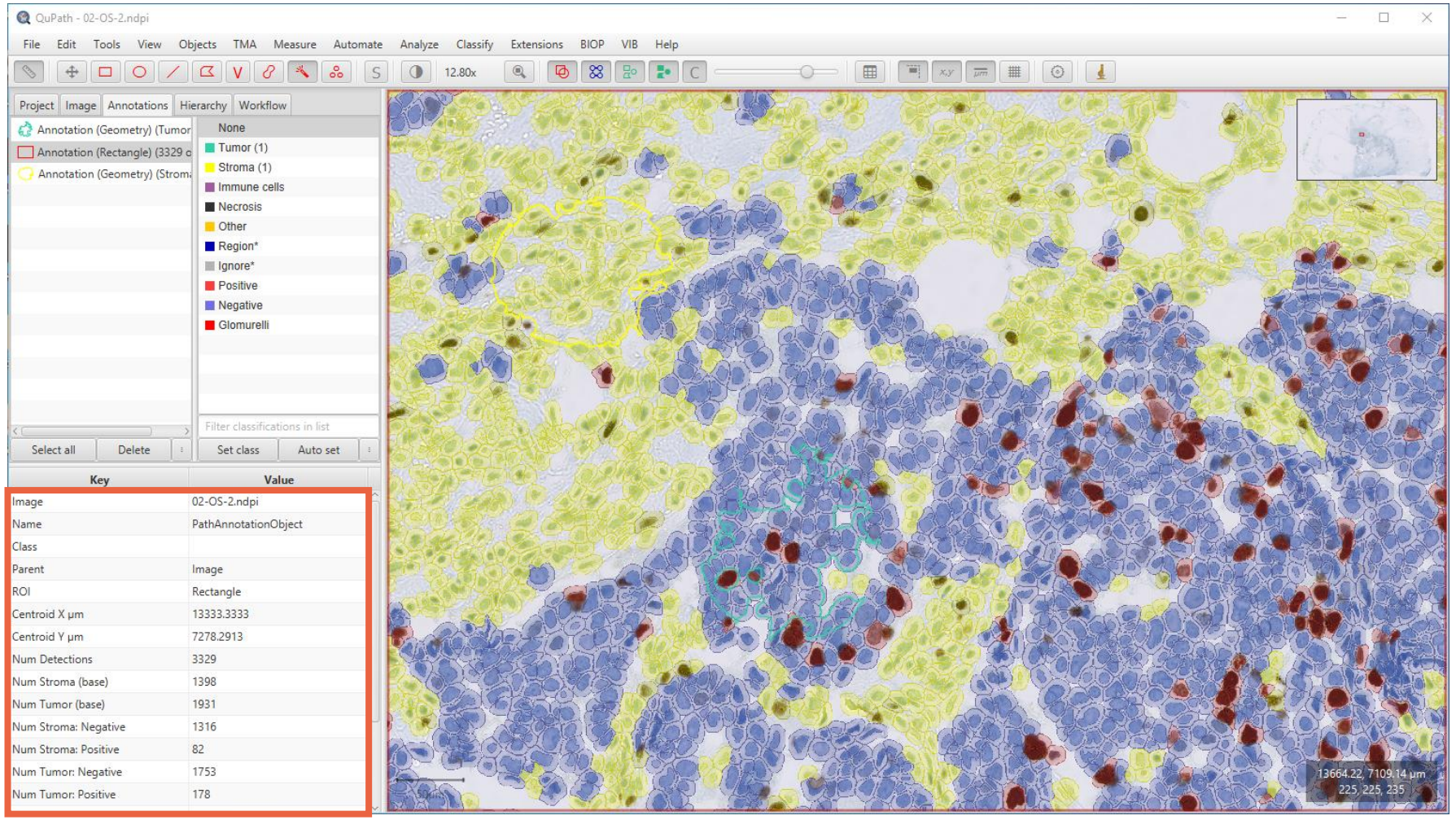
VIB

SCIENCE MEETS LIFE

QuPath : Train object classifier

[Classify>Object classification>Train Object classifier]

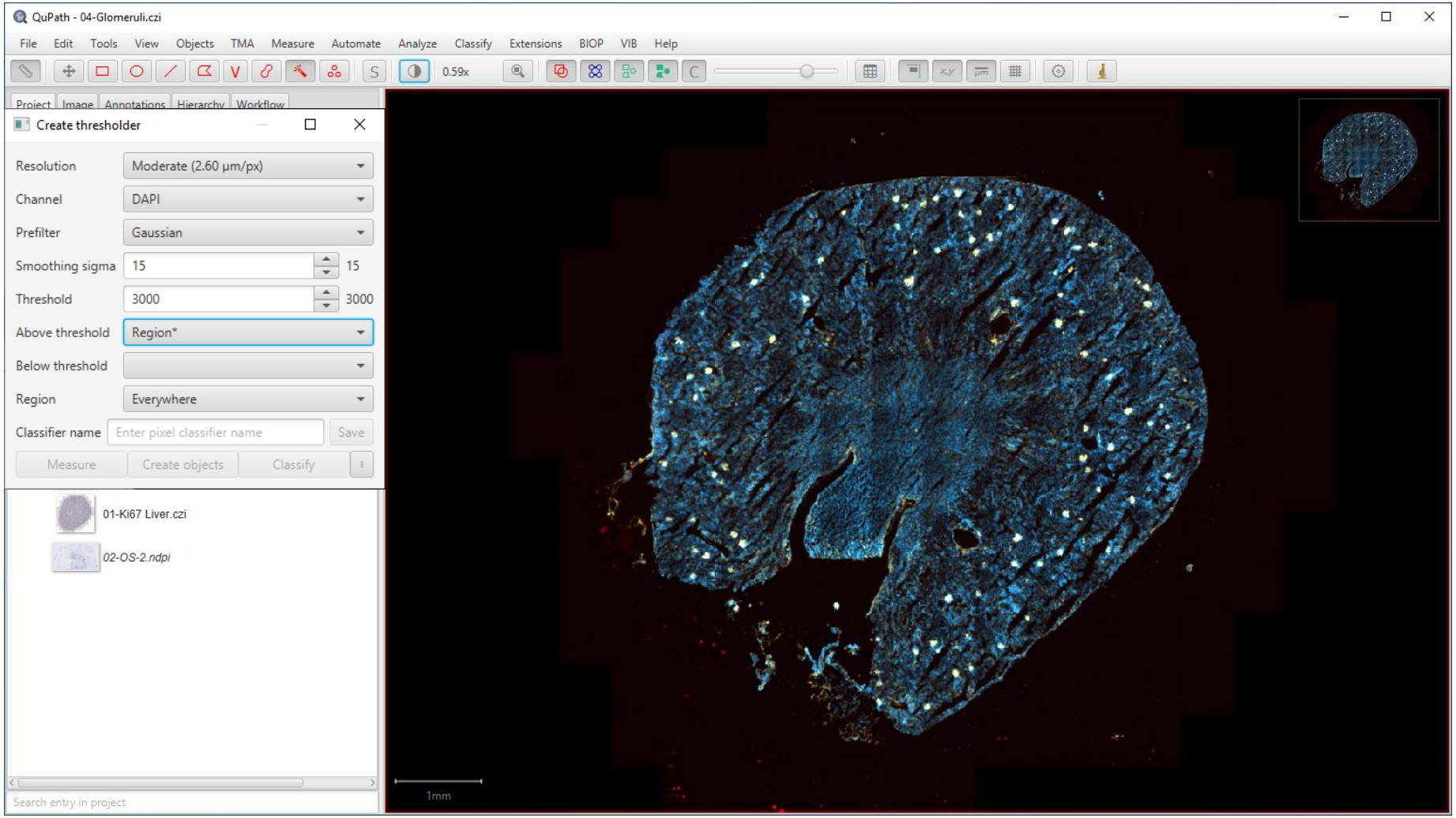
02-OS-2.ndpi



View the statistics
of the classification result
by annotation

QuPath : Detect Tissue with thresholder

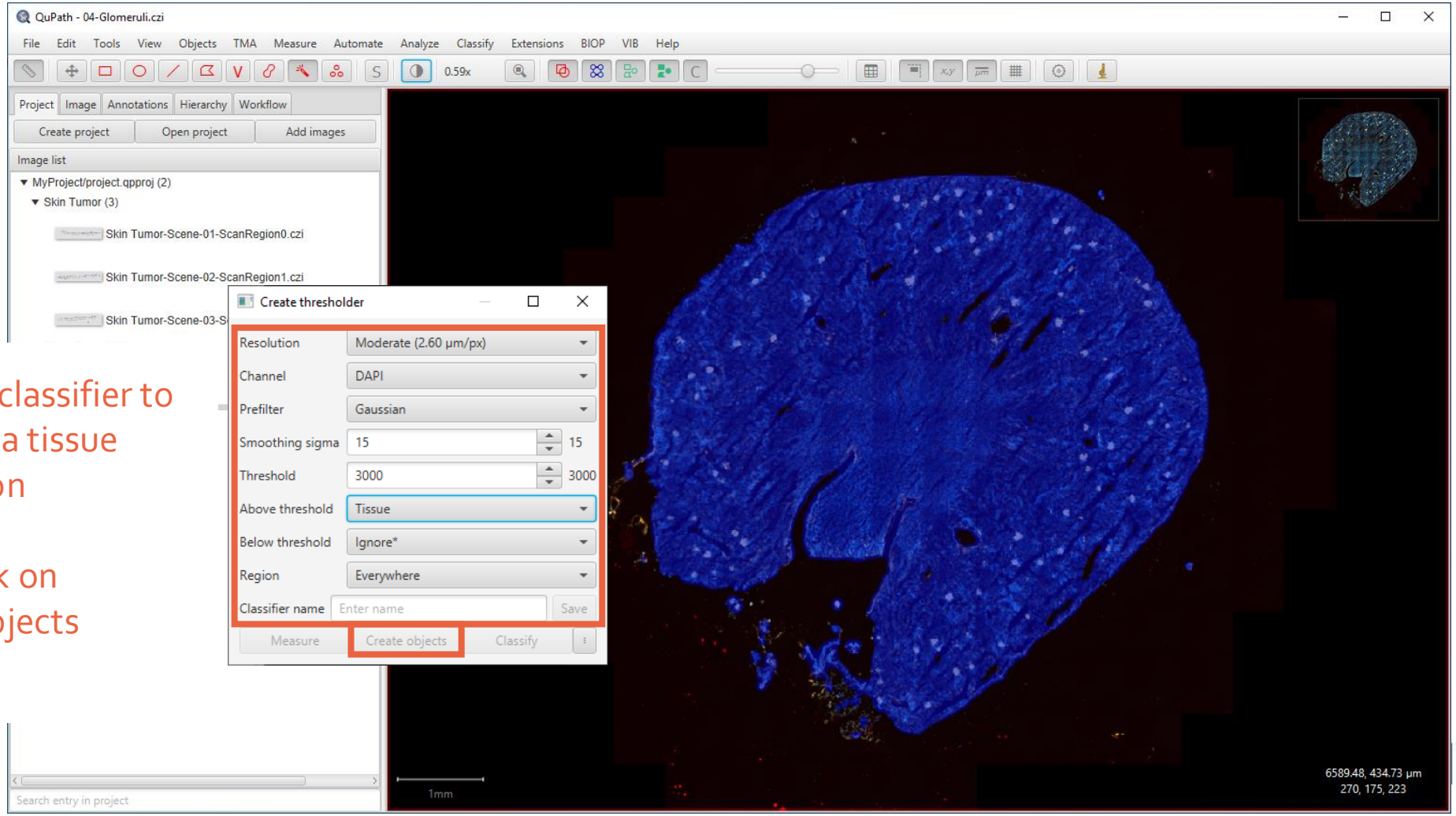
[Classify>Pixel classification>Create thresholder] 03-Glomeruli.czi



QuPath : Detect Tissue with thresholder

[Classify>Pixel classification>Create tresholder]

03-Glomeruli.czi



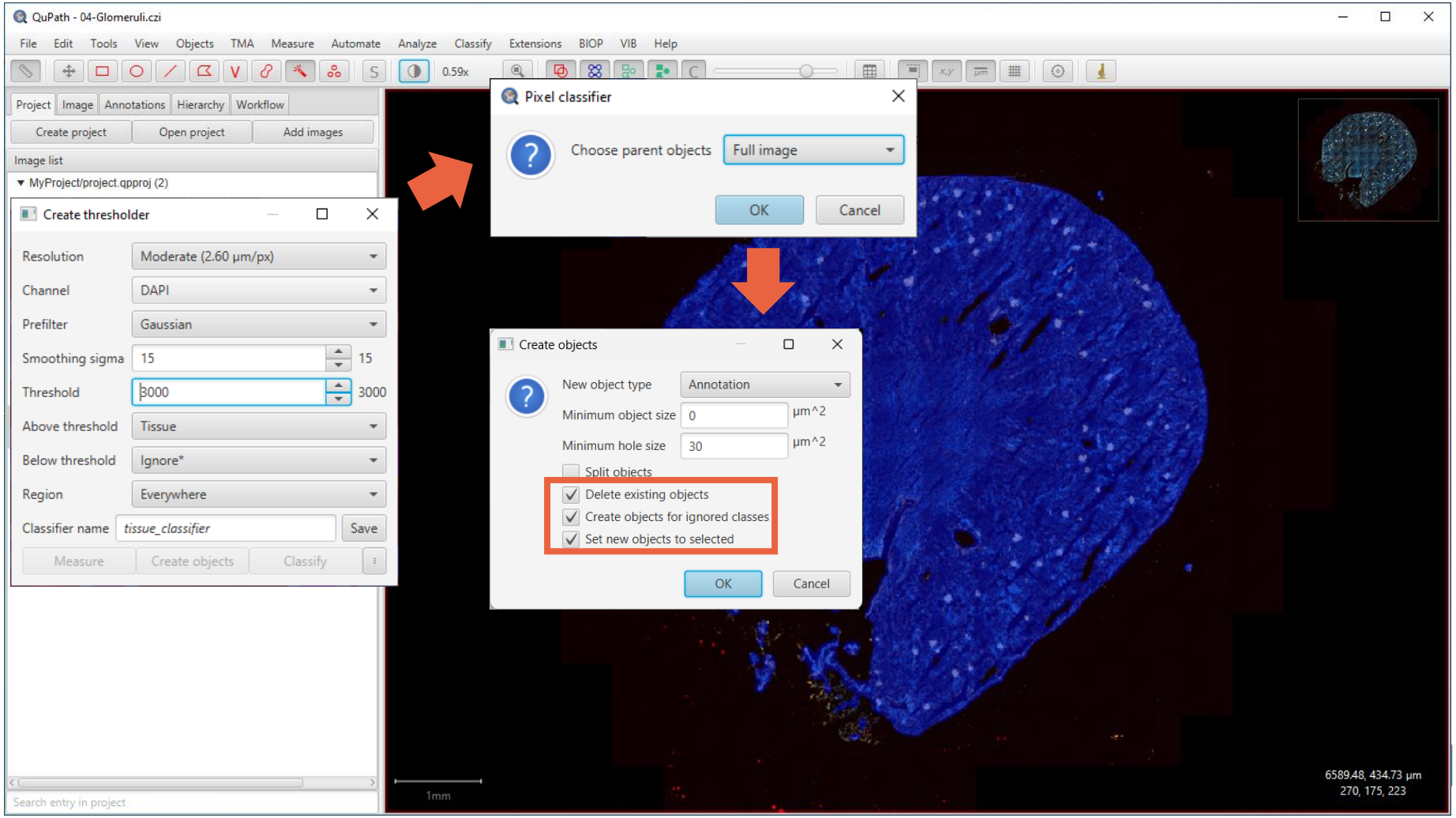
Save the classifier to to create a tissue annotation

Then click on Create objects

QuPath : Detect Tissue with thresholder

[Classify>Pixel classification>Create tresholder]

03-Glomeruli.czi



VIB

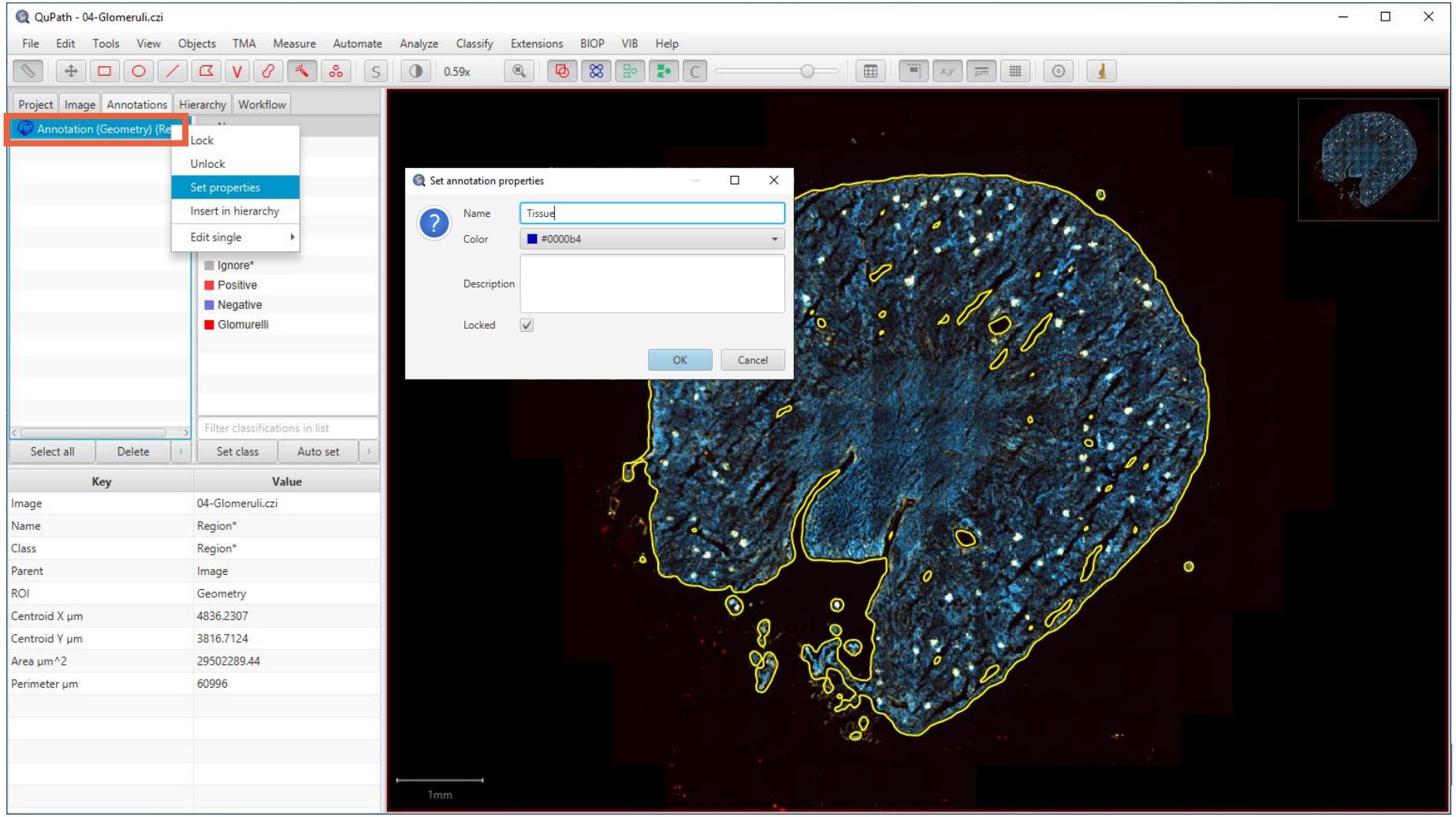
ENCE MEETS LIFE

QuPath : Detect Tissue with thresholder

[Classify>Pixel classification>Create tresholder]

03-Glomeruli.czi

Right click to rename the new annotation



VIB

ENCE MEETS LIFE

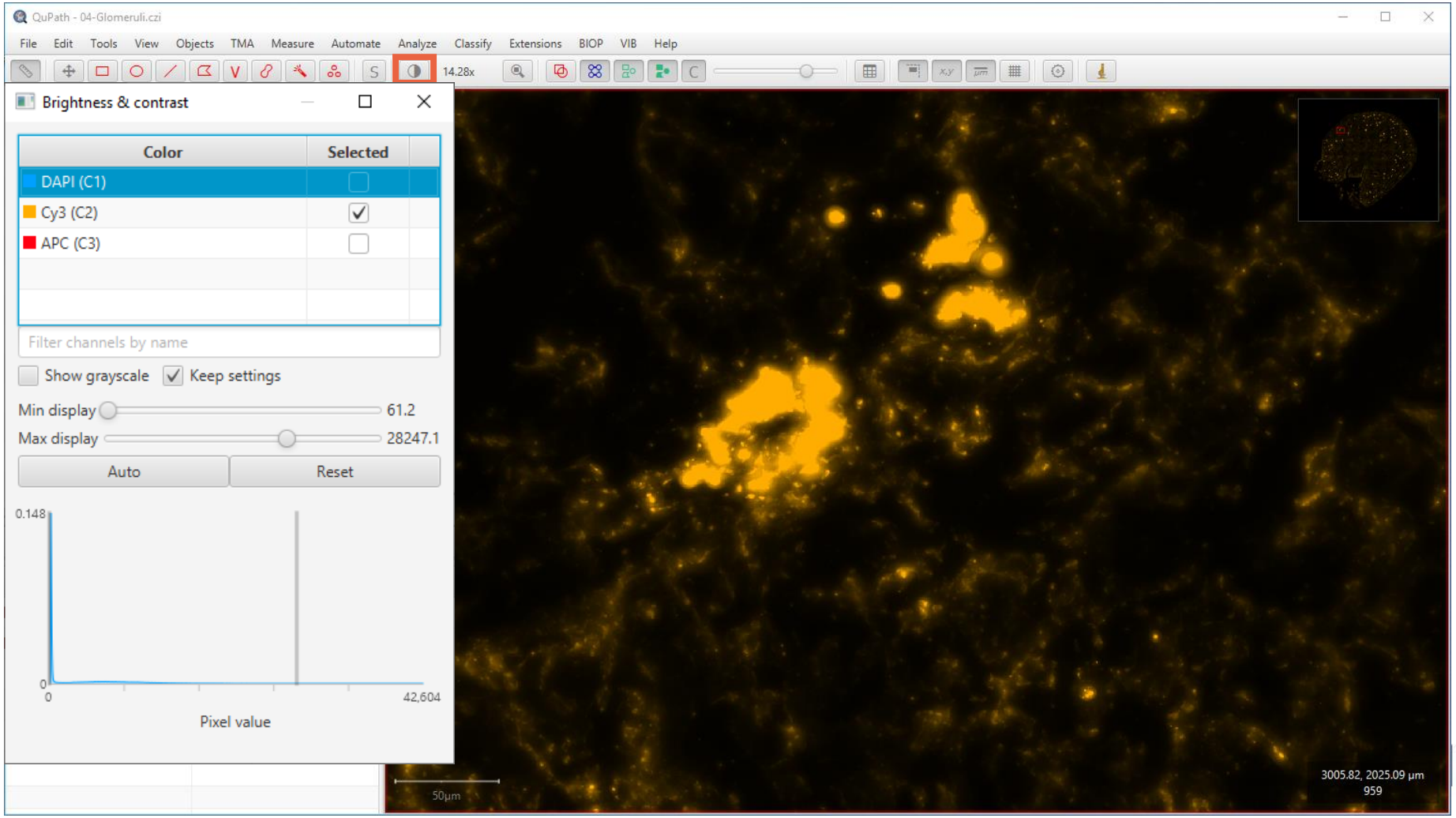
QuPath : Detect using pixel classifier

Create the annotation on the channel(s) of interest



03-Glomeruli.czi

Display only Glomerulli channel



VIB

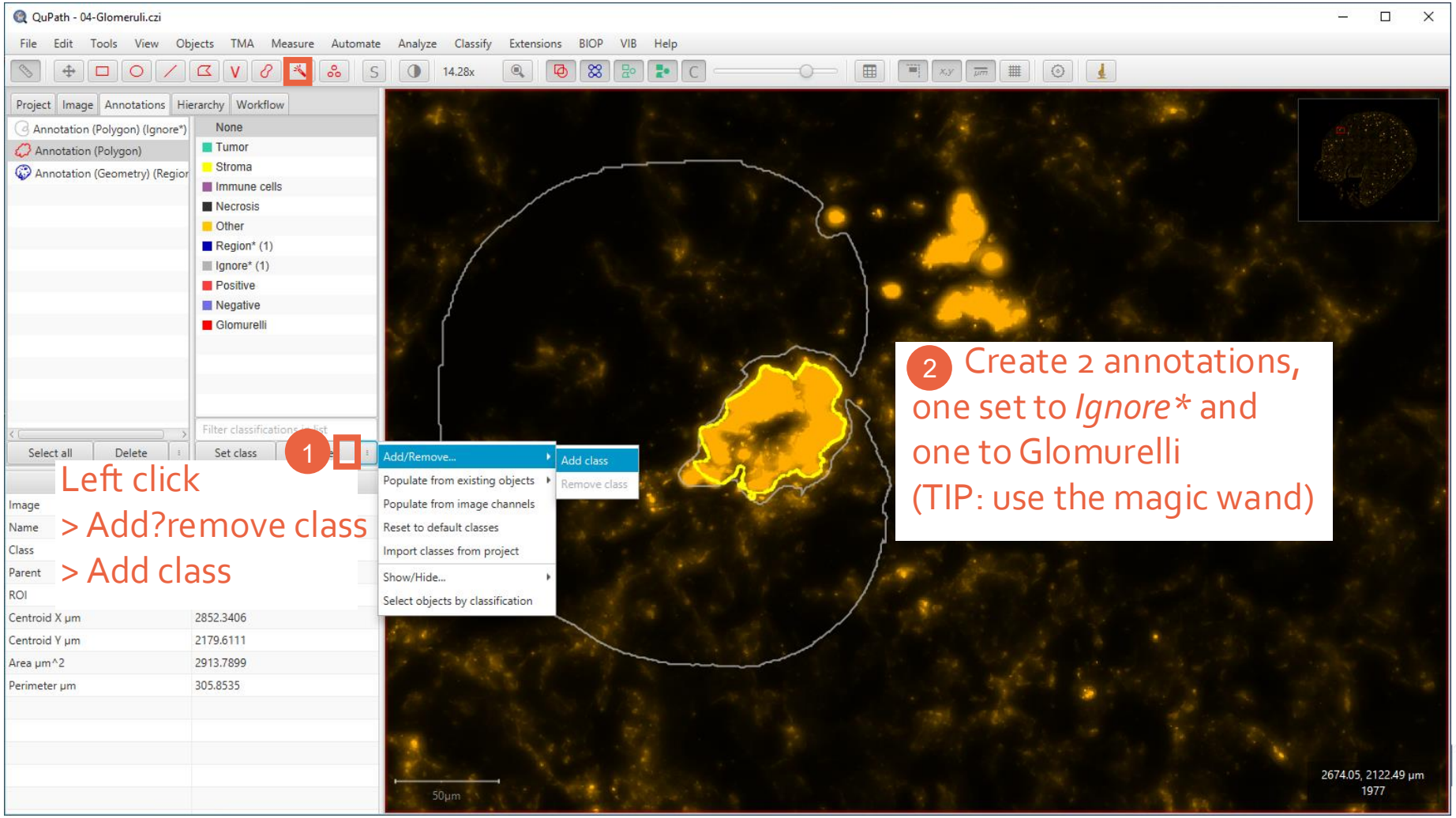
ENCE MEETS LIFE

QuPath : Detect with pixel classifier

Create a new class if needed and annotate



03-Glomeruli.czi



Avoid overlapping with an other annotation using **ctrl+shift+Brush (Command+shift+Brush)**

VIB

ENCE MEETS LIFE

QuPath : Detect Tissue with pixel classifier

[Classify>Pixel classification>Train pixel classifier]

03-Glomeruli.czi

4 Select the tissue annotation

7 Create objects

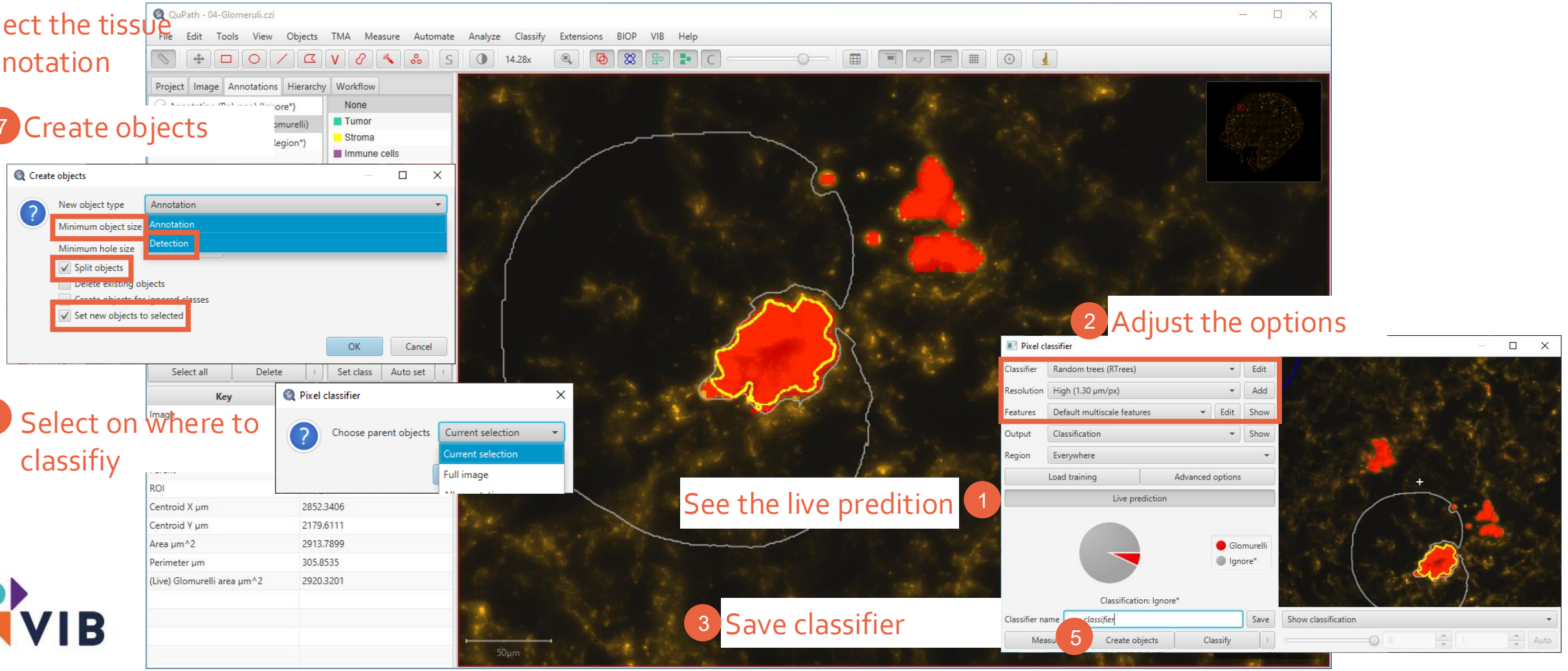
6 Select on where to classifiy

2 Adjust the options

See the live predition 1

3 Save classifier

5

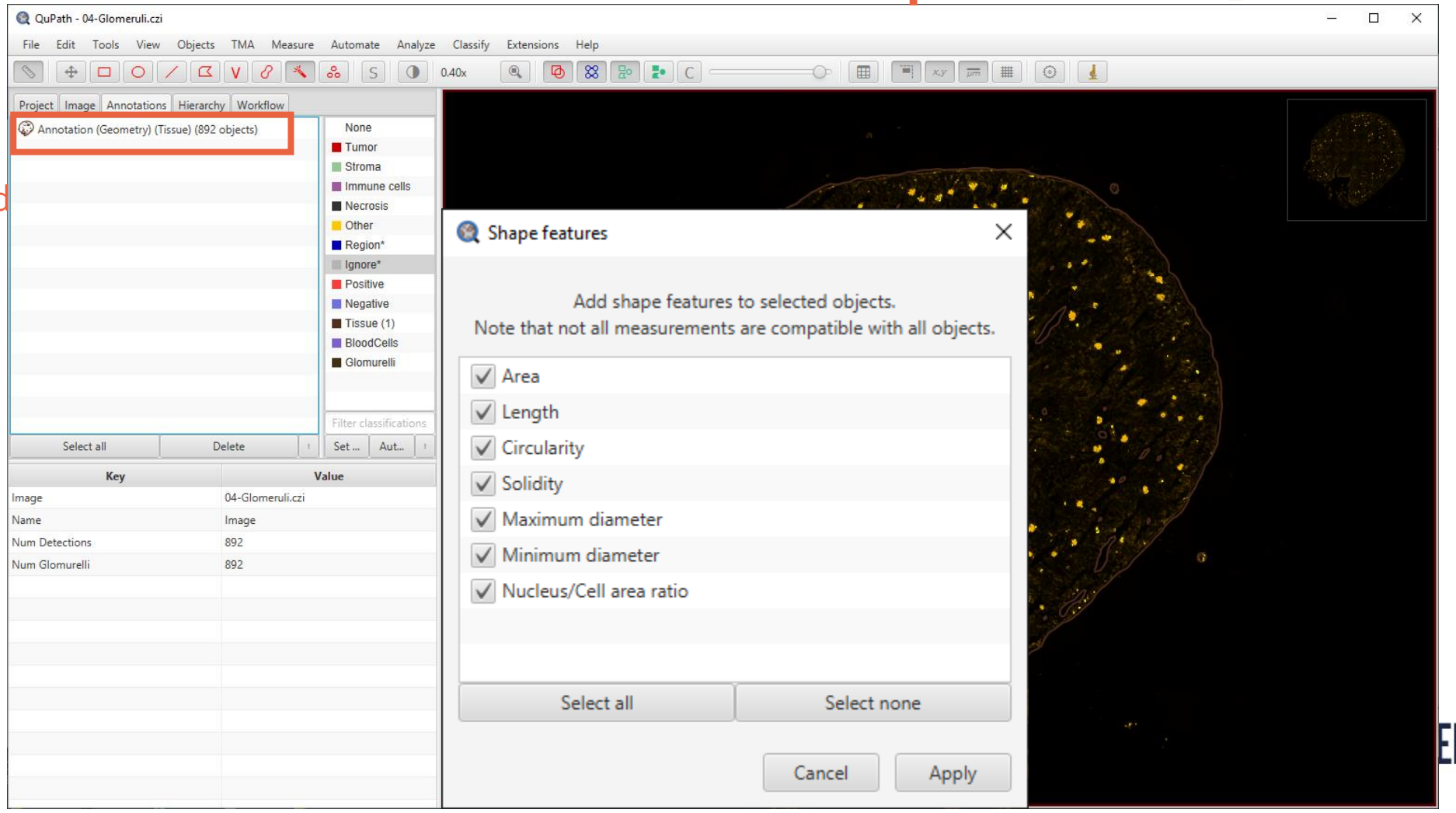


QuPath : Add extra features

[Objects>Select…>Select detections…>Select all detection]

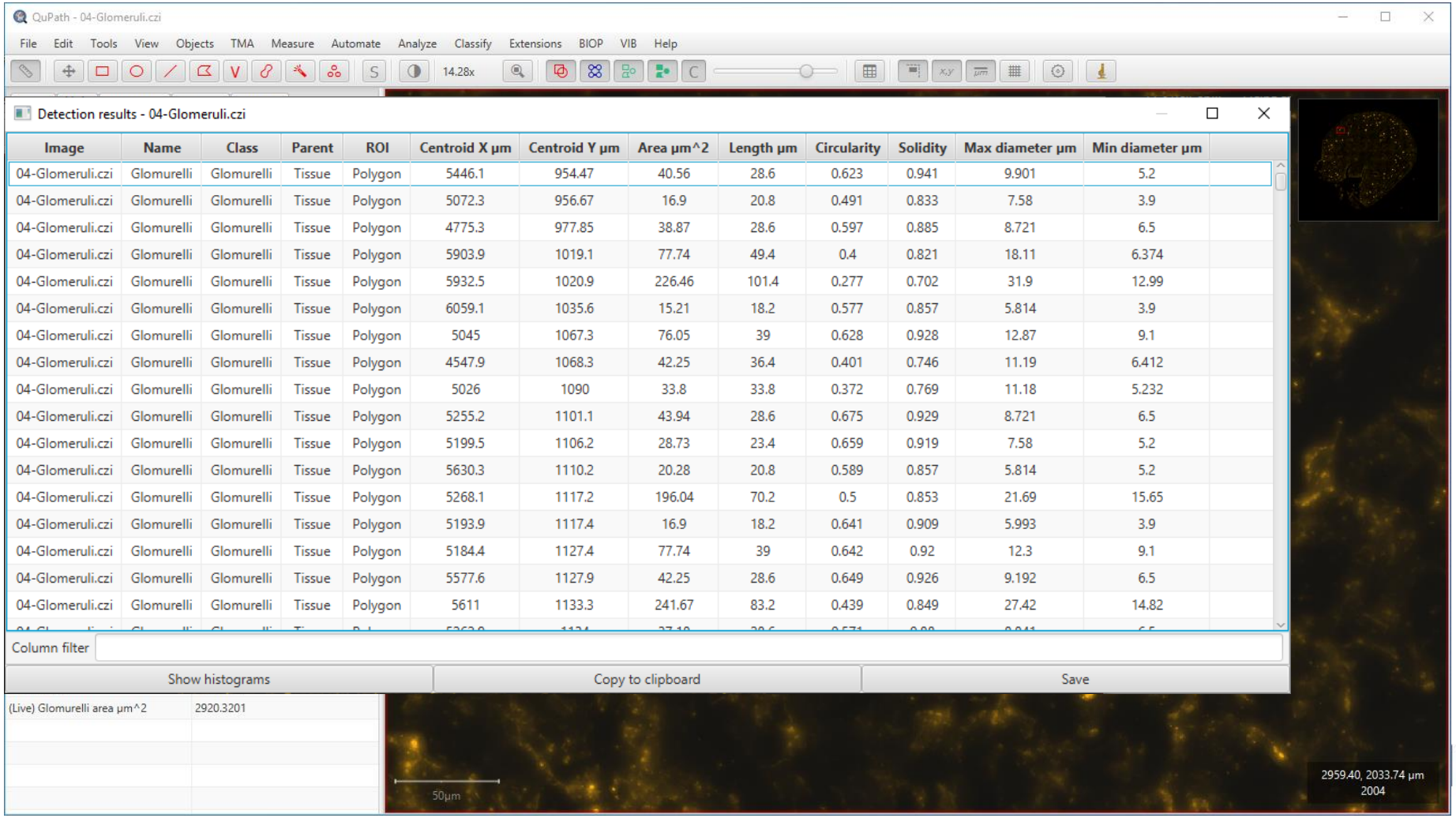
[Analyze>Calculate Features>Add shape features]

See the created detections in the processed annotation



QuPath : View the measuremement

[Measure>Show Detections Measurements]

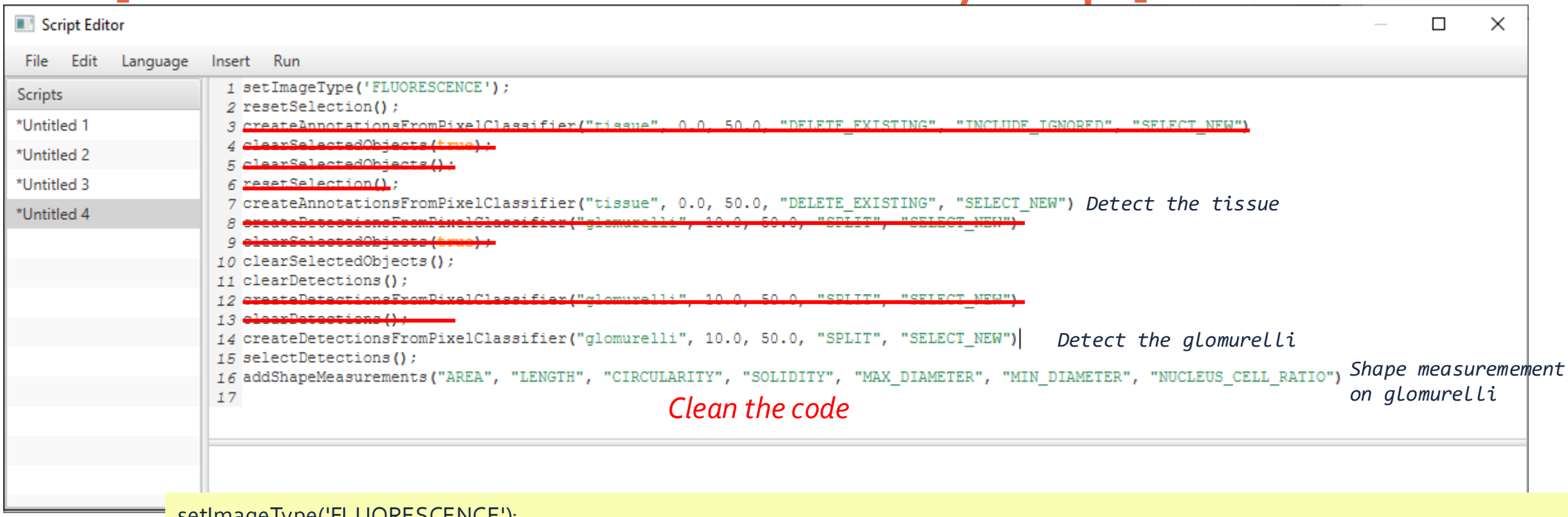


VIB

ENCE MEETS LIFE

QuPath : Create a script

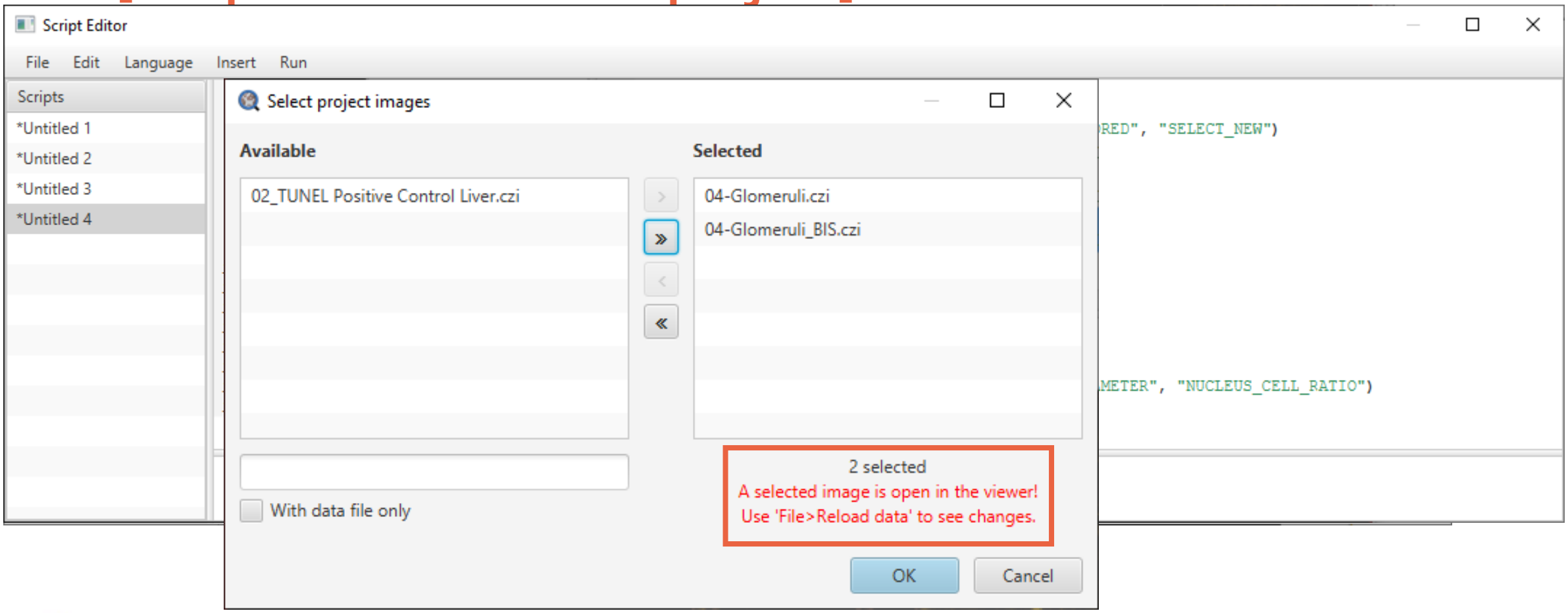
[Automate>Create command history Script]




```
setImageType('FLUORESCENCE');
resetSelection();
createAnnotationsFromPixelClassifier("tissue", 0.0, 50.0, "DELETE_EXISTING", "SELECT_NEW")
clearDetections();
createDetectionsFromPixelClassifier("glomurelli", 10.0, 50.0, "SPLIT", "SELECT_NEW")
selectDetections();
addShapeMeasurements("AREA", "LENGTH", "CIRCULARITY", "SOLIDITY", "MAX_DIAMETER", "MIN_DIAMETER", "NUCLEUS_CELL_RATIO")
```

QuPath : Run the script in batch

[Script Editor>Run For project]



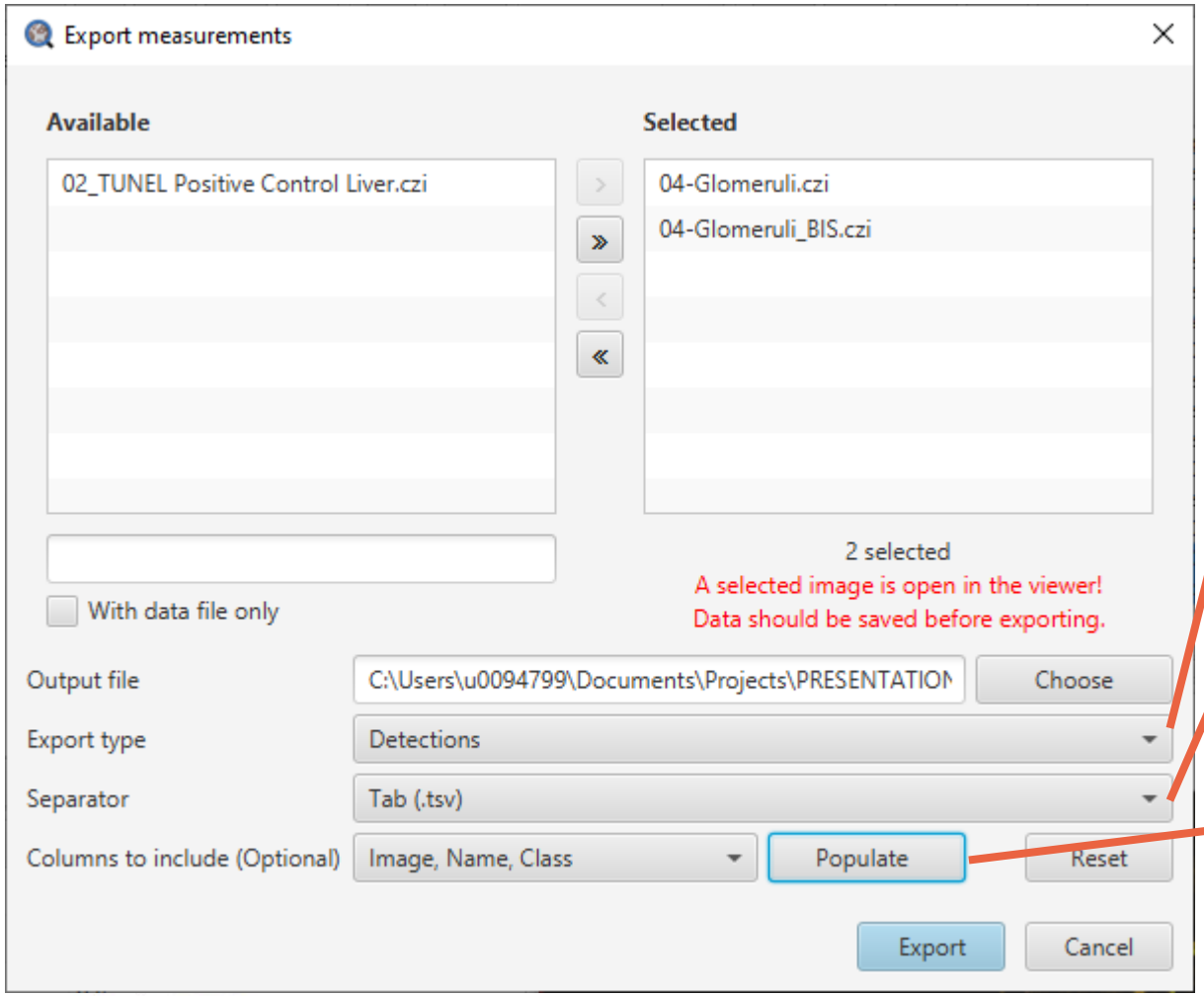
Make sure the image you want to analyze are not open

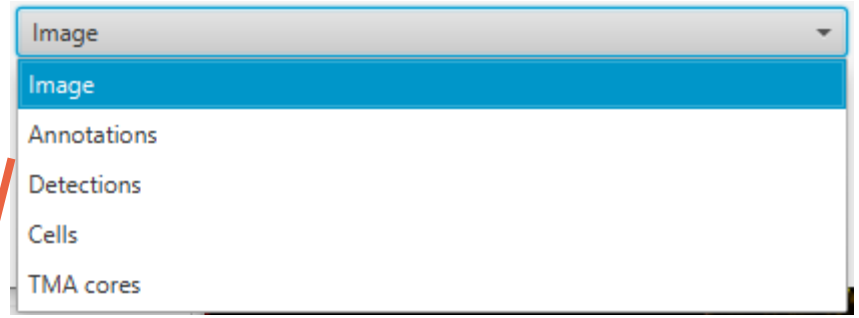
VIB

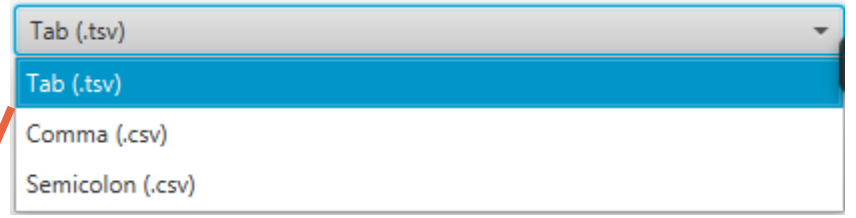
SCIENCE MEETS LIFE

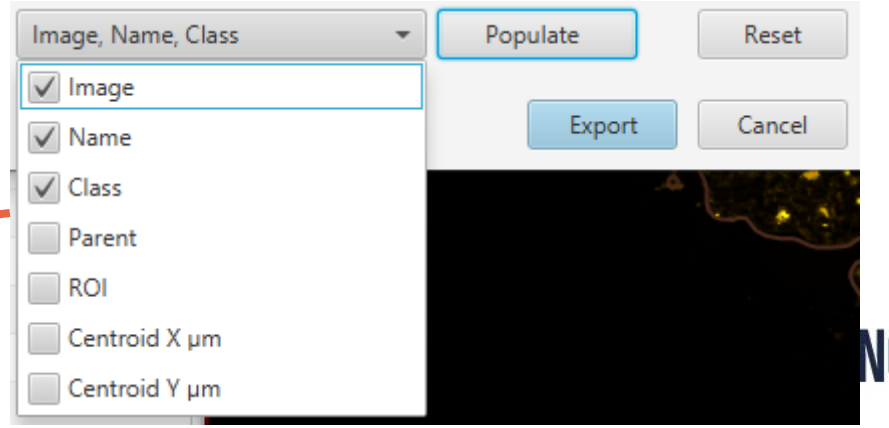
QuPath : Export all measurements

[Measure>Export measurements]





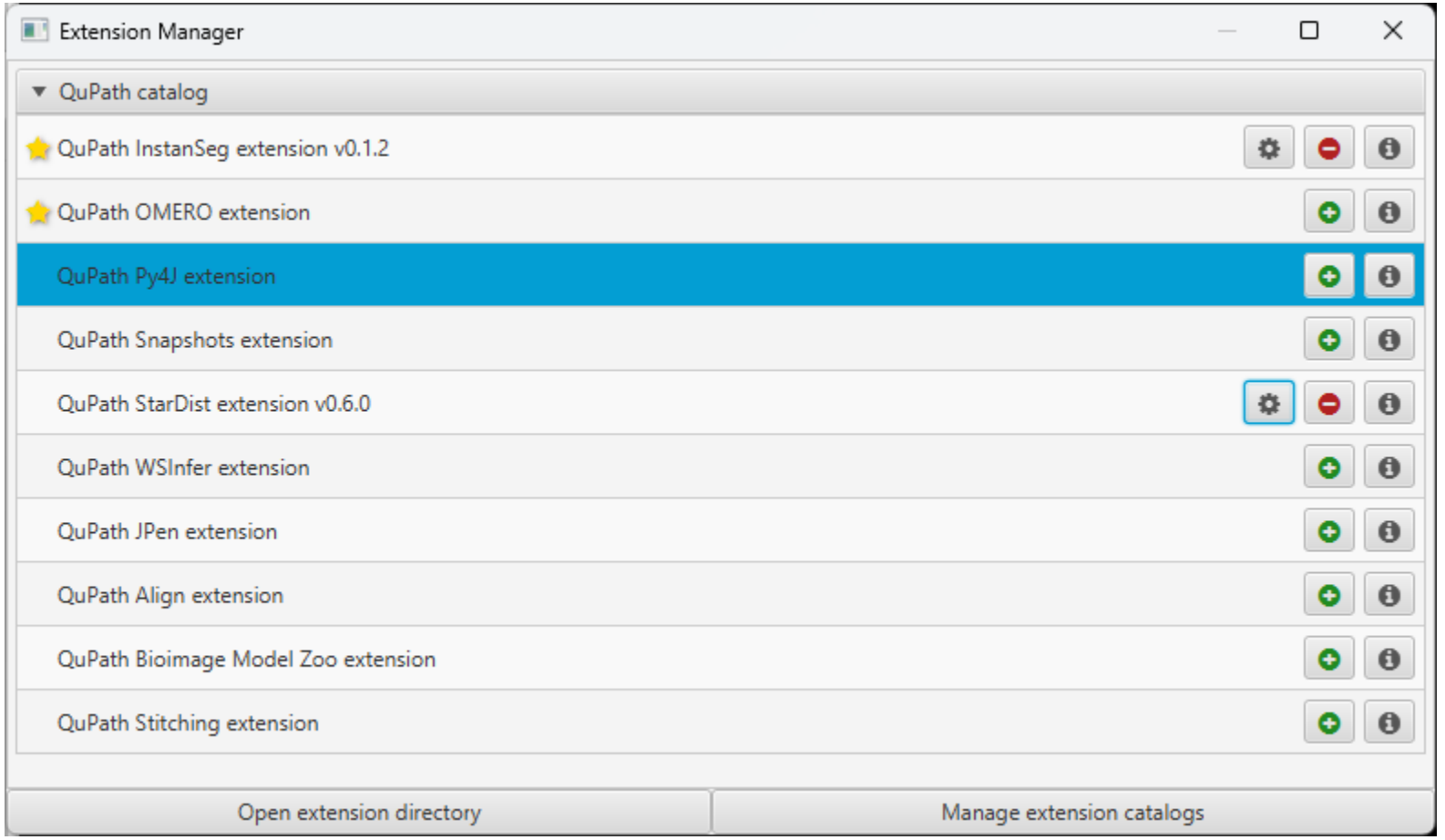




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QuPath : Extensions

[Extensions>Manage Extensions]

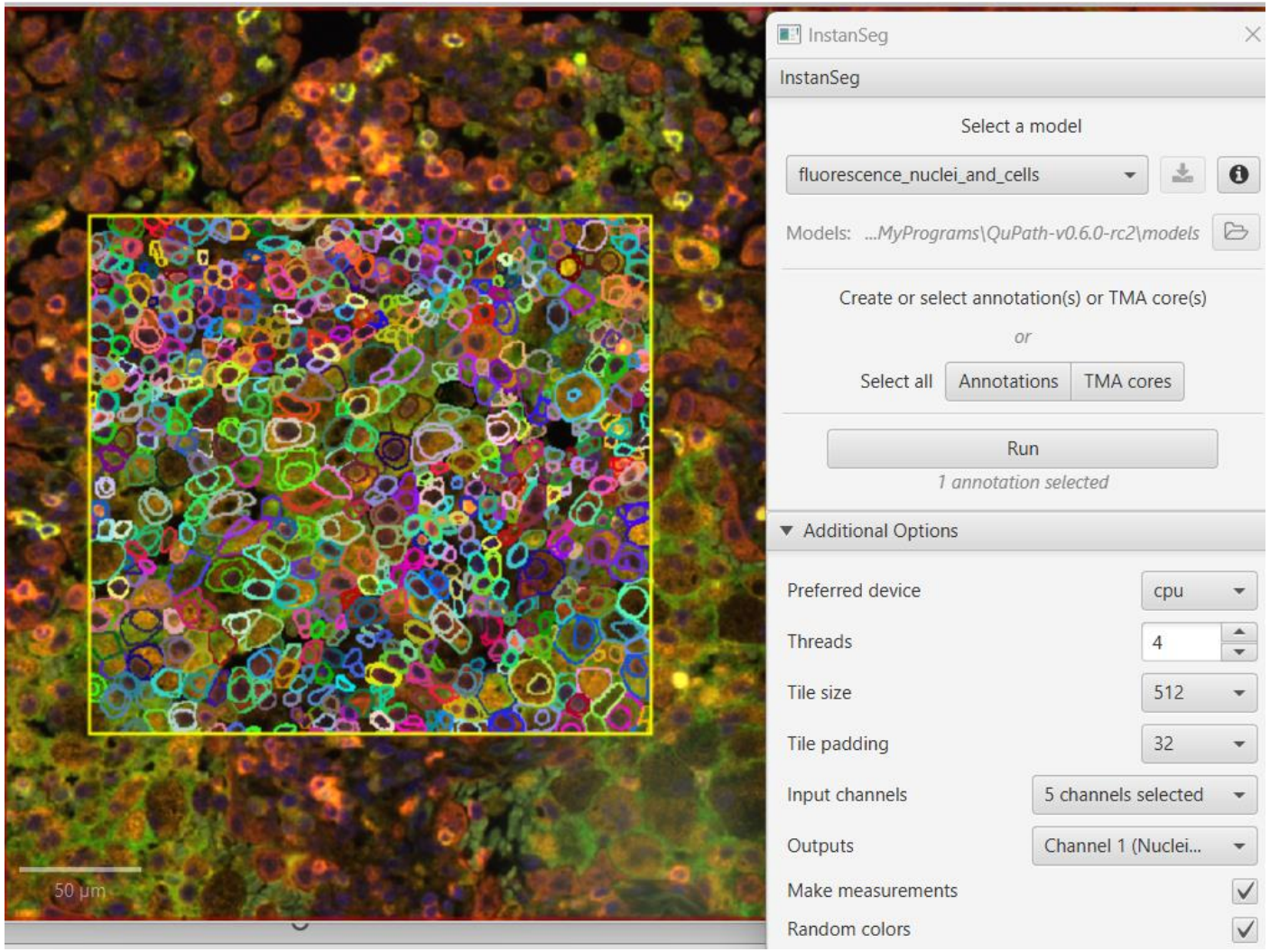


VIB

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QuPath with Instanseg

try it



- Included in qupath as plugin
- No need of conda/python
- Segment nuclei AND cells
- Work on CPU only

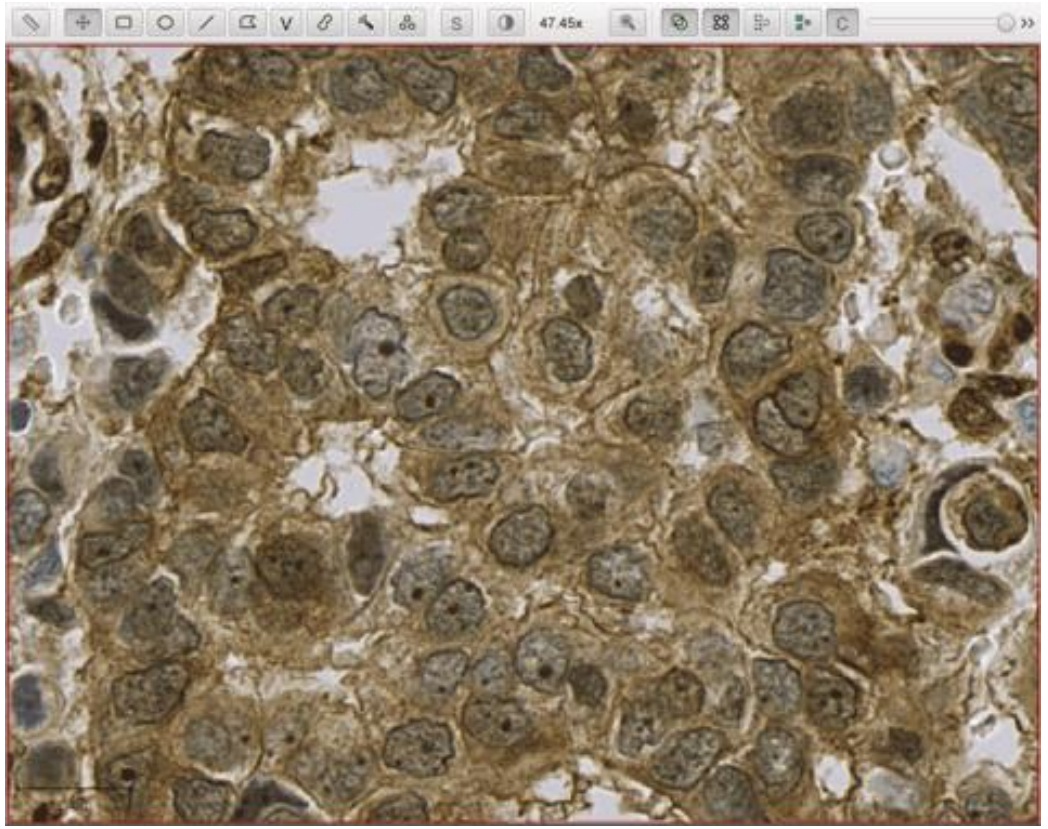
- Github
- Publication

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QuPath Extension : StarDist

Segment nuclei using a dedicated neural network

- Documentation : https://qupath.readthedocs.io/en/0.5/docs/deep/stardist.html#stardist-extension



QuPath Links



Links

- Documentation :https://qupath.readthedocs.io/en/latest/
- Forum : https://forum.image.sc/tag/qupath
- Videos : https://www.youtube.com/watch?v=4An5n6Y_rRI
  https://www.youtube.com/@zbigniewmikulski/videos
- Script collection :https://gist.github.com/Svidro

image.sc

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